ACADEMIC REGULATIONS & SYLLABUS

Faculty of Computer Science & Applications
Smt. Chandaben Mohanbhai Patel Institute of Computer Applications
MCA Programme

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Computer Science & Applications

ACADEMIC REGULATIONS

Master of Computer Application Programme

(Choice Based Credit System)

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand
Phone: 02697-247500, Fax: 02697-247100, Email: info@charusat.ac.in
www.charusat.ac.in

Year – 2019-2020

CHARUSAT

FACULTY OF COMPUTER SCIENCE AND APPLICATIONS
ACADEMIC RULES
Master of Computer Applications (MCA) Programme

To ensure uniform system of education, duration of post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

The Semester system of education should be followed across the Charotar University of Science and Technology (CHARUSAT) at Master's levels. Each semester will be at least 90 working days duration. Every enrolled student will be required to take a specified load of course work in the chosen course of specialization and also complete a project/dissertation if any.

2. Duration of Programme

Postgraduate programme : Master of Computer Applications

Minimum	4 semesters (2 academic years)
Maximum	6 semesters (3 academic years)

3. Eligibility & Mode of admissions

Eligibility of a candidate and mode of admission to the programme will be according to the regulations for admission committee decided by Government of Gujarat from time to time.

4. Programme structure and Credits

A student admitted to a program should study the course and earn credits specified in the course structure.

5. Attendance

5.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

5.2 Student attendance in a course should be 80%.

6 Course Evaluation

6.1 The performance of every student in each course will be evaluated as follows:

6.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and

6.1.2 Final examination by the University through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, for 70% of the marks for the course.

6.2 *University Examination*

6.2.1 The final examination by the University for 70% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these.

6.2.2 In order to earn the credit in a course a student has to obtain grade other than FF.

6.3 Performance at Internal & University Examination will be done on the relative grading system.

7 Grading

The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are defined as follows:

SGPA = $\Sigma C_i G_i / \Sigma C_i$ where C_i is the number of credits of course i
G_i is the Grade Point for the course i
and i = 1 to n, n = number of courses in the semester

CGPA = $\Sigma C_i G_i / \Sigma C_i$ where C_i is the number of credits of course i
G_i is the Grade Point for the course i
and i = 1 to n, n = number of courses of all semesters up to which CGPA is computed.

8. Detention Rule

8.1 No student will be allowed to move further in next semester if CGPA is less than 3 at the end of an academic year..

8.2 A Student will not be allowed to move to third year if he/she has not cleared all the courses of first year.

8.3 A student will not be allowed to move to fourth year if he/she has not cleared all the courses of first and second year.

9. Awards of Degree

9.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

9.1.1 He should have earned at least minimum required credits as prescribed in course structure; and

9.1.2 He should have cleared all evaluation components in every course; and

9.2 The student who fails to satisfy minimum requirement of CGPA will be allowed to improve the grades so as to secure a minimum CGPA for the award of degree. Only latest grade will be considered.

10 Award of Class:

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction:	CGPA ≥7.5 & ≤10
First class:	CGPA ≥ 6.0 & <7.5
Second Class:	CGPA ≥ 5.0 & <6.0
Pass Class:	CGPA <5.0

11 Transcript:

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)

FACULTY OF COMPUTER SCIENCE & APPLICATIONS

CHOICE BASED CREDIT SYSTEM

FOR

MASTER OF COMPUTER APPLICATIONS

CHOICE BASED CREDIT SYSTEM

The choice based credit system provides flexibility in designing curriculum and assigning credits based on the course content and hour of teaching. The choice based credit system provides an opportunity for the students to choose courses from the prescribed courses comprising core, elective and open elective courses. The CBCS provides a cafeteria type approach in which the students can take courses of their choice and adopt an interdisciplinary approach to learning. The courses shall be evaluated on the grading system, which is considered to be better than the conventional marks system.

CBCS – Conceptual Definitions / Key Terms (Terminologies)

Types of Courses: The Programme Structure consist of 3 types of courses:

Foundation Courses, Core Courses and Elective Courses.

Foundation Course

These courses are offered by the institute in order to prepare students for studying courses to be offered at higher levels.

Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

A. University Core Courses(UC):

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme.

B. Programme Core Courses(PC):

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialized or advanced or supportive to the discipline of study or which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidates proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

A. University Elective Courses(UE):
The pool of elective courses offered across all faculties / programmes.

B. Institute Elective Course (IE)
Institute elective courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialization.

C. Programme Elective Courses(PE):
The programme specific pool of elective courses offered by respective programme.

Vision

To become a leading institution in the field of computer applications and contribute in national efforts of computerizing public systems

Mission

To produce competent computer professionals with the ability to face future challenges.

Program Educational Objectives (PEOs)

Graduates of Master of Computer Applications(MCA) program at the Charotar University of Science and Technology (CHARUSAT) should be able to utilize the knowledge gained from their academic program to :

- PEO1 Be successful professionals in industry, government, academia, research, entrepreneurial pursuit and consulting firms.
- PEO2 Integrate fundamentals and advanced concepts of computer science and applications to develop innovative solutions to societal problems.
- PEO3Function and communicate effectively, both individually and within multidisciplinary teams.
- PEO4 Pursue life-long learning to fulfill their goals.
- PEO5 Be sensitive to the consequences of their work, both ethically and professionally, for productive professional careers.

Program Outcomes (POs)

The curriculum of Master of Computer Applications(MCA) program at the Charotar University of Science and Technology (CHARUSAT) is designed to assure students, at the time of graduation are able to:

- PO1 Identify, critically analyze, formulate and develop computer applications.
- PO2 Design a computing system to meet desired needs within realistic constraint.
- PO3 Use modern computing tools and techniques for effective software engineering.
- PO4 Devise and conduct experiments, interpret data and provide well informed conclusions.
- PO5 Have a fundamental capability in oral and written communication.
- PO6 Function effectively within multidisciplinary teams.
- PO7 Understand the impact of system solutions in a contemporary and societal context for sustainable development.
- PO8 Function professionally with ethical responsibility.
- PO9 Appreciate the importance of goal setting and to recognize the need for life-long learning.

TEACHING SCHEME
FOR

MCA PROGRAMME
(1ST & 2ND YEAR)

EFFECTIVE FROM
ACADEMIC YEAR 2019-20

Teaching and Examination Scheme (MCA Programme)
Effective from Year 2019-20

Semester-I

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA843	Enterprise Computing using Java EE	3	3	6	6	10	20	70	15	15	70	200
CA844	Visual Programming	3	3	6	6	10	20	70	15	15	70	200
CA845	Advanced Database Technologies	3	3	6	6	10	20	70	15	15	70	200
CA846	Software Engineering Paradigms	3	-	3	3	10	20	70	-	-	-	100
CA847-CA850	Elective-I	3	-	3	3	10	20	70	-	-	-	100
HS703.01C	Languages (French)	-	2	2	2	-		-	30		70	100
HS704 C	Academic Speaking	-	2	2	2	-		-	30		70	100
	University Elective-I **	-	2	2	2	-	-	-	30		70	100
		15	13	28	28	500			500			1000

** Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **CA730 Internet and Web Designing** course for others.

Elective-I

1. CA847 Cloud Computing
2. CA848 Cyber Security and Computer Forensics
3. CA849 Internetworking with TCP/IP Protocol Suite
4. CA850 Advanced Operating Systems

University Elective-I

No	Course Code	Course Name	Department/Faculty
1	MA771.01	Reliability and Risk Analysis	Mathematics
2	EE781.01	Optimization Techniques	Engineering
3	ME781.01	Occupational Health & Safety	Engineering
4	CE772.01	Research Methodology	Engineering
5	PT795.01	Health & Physical Activity	Physiotherapy
6	NR755	First Aid & Life Support	Nursing
7	RD701.01	Introduction to Analytical Techniques	Applied Science
8	RD702.01	Introduction to Nanoscience and Technology	Applied Science
9	MB650	Creative Leadership	Management
10	PH891	Community Pharmacy Ownership	Pharmacy
11	PD261	Astrophysics, Space and Cosmos-I	Applied Science

Teaching and Examination Scheme (MCA Programme)
Effective from Year 2019-20
Semester-II

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA851	Web Application Development using Open Source Technology	3	3	6	6	10	20	70	15	15	70	200
CA852	Software Quality Assurance	3	3	6	6	10	20	70	15	15	70	200
CA853	Responsive Web Designing Using Framework	-	2	2	2	-	-	-	15	15	70	100
CA854-CA855	**Elective II**	3	3	6	6	10	20	70	15	15	70	200
CA856-CA858	**Elective-III**	3	3	6	6	10	20	70	15	15	70	200
HS705 C	Academic Writing	-	2	2	2	-	-	-	30		70	100
	University Elective-II **	-	2	2	2	-	-	-	30		70	100
		12	18	30	30	400			700			1100

** Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **CA842 Mobile Application Development** course for others.

Elective-II
1. CA854 Mobile Application Technology – Android Platform
2. CA855 Mobile Application Development-iOS Platform

Elective-III
1. CA856 HTTP Web Service for Enterprise Application
2. CA857 Framework and Applications
3. CA858 Multi Paradigm Scripting using Python

University Elective-II

No	Course Code	Name	Department/Faculty
1	EE782.01	Energy Audit and Management	Engineering
2	CE771.01	Project Management	Engineering
3	PT796.01	Fitness & Nutrition	Physiotherapy
4	NR752.01	Epidemiology and Community Health	Nursing
5	OC733.01	Introduction to Polymer Science	Applied Science
6	MB651	Software based Statistical Analysis	Management
7	PH892	Intellectual Property Rights	Pharmacy
8	MA772.01	Design of Experiments	Mathematics
9	PD262	Astrophysics, Space and Cosmos-II	Applied Science

Teaching and Examination Scheme (MCA Programme)

Effective from Year 2019-20

Semester-III

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA926	Open Source Frameworks	3	3	6	6	10	20	70	15	15	70	200
CA927	Data Analytics	3	3	6	6	10	20	70	15	15	70	200
CA928-CA931	Elective-IV	3	-	3	3	10	20	70	-	-	-	100
CA932	**Mini Project**	-	15	15	15	-	-	-	100		200	300
		9	21	30	30	300			500			800

Elective-IV

1. CA928 Artificial Intelligence
2. CA929 Digital Image Processing
3. CA930 Compiler Construction
4. CA931 Analysis and Design of Algorithms

Teaching and Examination Scheme (MCA Programme)
Effective from Year 2019-20

Semester-IV

Course Code	Course Title	Teaching Scheme				Internal	End Semester Examination		Total
		Contact Hours			Credit	Continuous Evaluation	Report	Presentation & Viva	
		Inst.	Industry	Total					
CA933	Project Work	2	28	30	30	200	200	400	800

OBJECTIVES, TEACHING SCHEME & DETAILED SYLLABUS

FOR

MCA PROGRAMME
(1st SEMESTER)

EFFECTIVE FROM
ACADEMIC YEAR 2019-20

Teaching and Examination Scheme (MCA Programme)
Effective from Year 2019-20

Semester-I

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA843	Enterprise Computing using Java EE	3	3	6	6	10	20	70	15	15	70	200
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CA846	Software Engineering Paradigms	3	-	3	3	10	20	70	-	-	-	100
CA847-CA850	Elective-I	3	-	3	3	10	20	70	-	-	-	100
HS703.01C	Languages (French)	-	2	2	2	-		-	30		70	100
HS704 C	Academic Speaking	-	2	2	2	-		-	30		70	100
	University Elective-I **	-	2	2	2	-	-	-	30		70	100
		15	13	28	28	500			500			1000

****** Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **CA730 Internet and Web Designing** course for others**.**

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3. CA849 Internetworking with TCP/IP Protocol Suite
4. CA850 Advanced Operating Systems

University Elective-I

No	Course Code	Course Name	Department/Faculty
1	MA771.01	Reliability and Risk Analysis	Mathematics
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3	ME781.01	Occupational Health & Safety	Engineering
4	CE772.01	Research Methodology	Engineering
5	PT795.01	Health & Physical Activity	Physiotherapy
6	NR755	First Aid & Life Support	Nursing
7	RD701.01	Introduction to Analytical Techniques	Applied Science
8	RD702.01	Introduction to Nanoscience and Technology	Applied Science
9	MB650	Creative Leadership	Management
10	PH891	Community Pharmacy Ownership	Pharmacy
11	PD261	Astrophysics, Space and Cosmos-I	Applied Science

CA843 : Enterprise Computing using Java EE

(200 Marks)

Contact Hours: 06

Pre-requisite: Basic knowledge of Java Standard Edition concepts.

Methodology & Pedagogy: During lectures the more emphasis will be given on the fundamental knowledge of web application development and backend database management techniques using various topic given in the syllabus. During Practical sessions, students will be required to develop Applications using the concepts of JDBC, JSP, JSTL and Servlets. Student will also be explored to MVC architecture.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	Practical
1	Database Programming	07	36
2	Java Web Application Components	06	
3	Java Server Pages	06	
4	JSP Standard Tag Library	04	
5	Working with Servlets	07	
6	Advanced Servlet Features & Security	06	

Total Hours (Theory): 36

Total Hours (Practical): 36

Total: 72

Detailed Syllabus:

Unit – I: Database Programming Hours: 10

The Design of JDBC, The Structured Query Language, JDBC Configuration, Executing SQL Statements, Query Execution, Scrollable and Updatable Result Sets, Row Sets, Metadata, Transactions

Unit – II: Java Web Applications Components Hours: 09

Understanding Web Applications: Understanding Web Components, Servlets (Overview), JSP Pages, Introducing the MVC Design Pattern, Understanding Containers, Packaging Web Applications, Web Application Structure, JAR Files, WAR Files, HTTP, GET Request Method, POST Request Method, GET & POST in HTML Form Processing, Other Request Methods, The HTTP Response, Using Deployment Descriptors

Unit – III: Java Server Pages   Hours: 08

Introducing JSP Technology – Understanding the Page Life Cycle, JSP Documents, JSP Document Syntax, A Simple JSP Document, HTTP, Http Jsp Page Interface. Introducing JSP Syntax and Usage – General Rules of Syntax, JSP Elements, Available Object Scope, Implicit Objects, Scripting
Elements, Comments, JSP Expression Language

Unit – IV: Java Standard Tag Library   Hours: 05

Introduction to JSTL, Core Tag Library

Unit – V: Working with Servlets   Hours: 10

Working with Servlets: Introducing Servlet, Introducing Servlet & the MVC Pattern, Introducing javax.servlet Package, Introducing the Servlet Interface, Introducing the GenericServlet Class, Introducing HTTP & Servlets, Understanding the Request/ Response Cycle, Input & Output Streams, Introducing Servlet/ Container Communication, Introducing ServletContext,
Understanding the Deployment Descriptor, Introducing ServletContext Lifecycle Classes,
RequestDispatcher Interface, Using Filters & RequestDispatcher

Unit – VI: Advanced Servlet Features & Security   Hours: 06

Understanding the Stateless nature of HTTP, Why Track Client Identity & State? Maintain Sessions, Session Management Using the Servlet API, and Concepts of Filters.

Core Books:

1. Cay S Horstmann, Gary Cornell: Core Java, Volume II – Advanced Features, 8th Edition, Pearson Education.
2. Sue Spielman and Meeraj Kunnumpurath: Pro J2EE 1.4, Wiley Computer Publishing, 2004.
3. Marty Hall, Larry Brown: Core Servlets and JavaServer Pages, Volume 2, Advanced Technologies, 2nd Edition, Pearson Education, 2008.

Reference Books:

1. Bryan Basham, Kathy Sierra and Bert Bates: Head First Servlet and JSP, O'Reilly Publication, 1 st Edition.

Web References:

1. http://courses.coreservlets.com/Course-Materials/csajsp2.html [Servlet Basics]
2. http://www.ceit.es/asignaturas/InteInfo/Recursos/Servlets/JavaServlets.pdf [Servlet Tutorial PDF]
3. http://www.msuniv.ac.in/AdvancedJavaProgrammingwithDatabaseApplication.pdf [JDBC Tutorial]
4. www.doc.ic.ac.uk/~rcheung/teaching/2720/ppt/lecture12.ppt [JSP Tutorial Slides]

Course Outcomes: Upon successful completion of the course, the students will :

C01 : Able to perform various RDBMS connectivity using JDBC

C02 : Have enterprise application architecture and terminologies understanding

C03 : Able to develop web applications with database connectivity using Java Web Technologies

C04 : Be familiar with the concepts of JSPs, Servlets and related security issues.

C05 : Have knowledge regarding Java Standard Tag Library and Custom Tag creation.

Course Outcomes Mapping :

Unit No.	**Unit Name**	**Course Outcomes**				
		C01	**C02**	**C03**	**C04**	**C05**
1	Database Programming	✓		✓		
2	Java Web Applications Components		✓	✓		
3	Java Server Pages		✓	✓	✓	
4	Java Standard Tag Library					✓
5	Working with Servlets		✓	✓	✓	
6	Advanced Servlet Features & Security				✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	3	3	2	2	2	1	2	2
CO2	3	3	3	2	2	2	2	2	2
CO3	3	3	3	2	2	2	3	2	2
CO4	3	3	3	2	2	2	3	2	2
CO5	3	3	3	2	2	2	1	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA844 : Visual Programming
(200 Marks)

Contact Hours: 06

Pre-requisite: Familiarity with basic concepts of object oriented programming

Methodology & Pedagogy: During theory lectures illustrations emphasizing the need for advanced features of .Net framework and ASP.Net will be given. During Practical sessions, students will be required to develop Web Applications using concepts discussed during class.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	Practical
1	Introduction to .NET framework and ASP.NET	04	36
2	Working with ASP.NET Applications	12	
3	Database Connectivity through ADO.NET	06	
4	Overview of SOA	04	
5	Reporting	05	
6	Advanced ASP.NET	05	

Total Hours (Theory): 36
Total Hours (Lab): 36
Total: 72

Detailed Syllabus:

Unit - I : Introduction to .NET framework and ASP.NET Hours:04

- Introducing .NET Framework
  - Enterprise vision of .NET
  - .NET Framework Component
  - .NET Framework Version Compatibility
  - Core of .NET Framework: Application Services, Base Class Library and CLR
- Introducing Web Programming.
  - Understanding Web server (IIS) and Web Client.
  - Basic of Http Request and Http Response.
  - Understand form Tag and Comparison between Get and Post Methods
- Introducing ASP and ASP.NET
  - Programming in ASP using visual studio

- Programming in ASP.NET using visual studio
- Deploying ASP.NET Application
  - Compilation and execution of .NET Application
  - Dynamic Compilation

Unit - II: Working with ASP.NET Applications Hours:12

- ASP.NET Page Life Cycle
- Structure of an ASP.NET Page: ASPX Page, Code behind File, WebConfig and machine config
- Develop Web Form
  - Create User Interface using Standard Controls, Rich Web Control, Navigation Controls and Login Control.
  - Working with properties and events of controls o Validate User Input using Validation Controls.
  - Concept of custom control.
- Concept of Master Page and Nested Master Page
- State Management Techniques
  - Client side: View State, Hidden Field, Cookie
  - Server Side: Application State, Session, Output cache
- Application Tracing, Error Handling and Responding to Errors o Try Catch Final
  - Custom Error Page

Unit - III: Database connectivity through ADO.NET Hours:06

- Introduction and Evolution of ADO.NET.
- Binding data to web controls and data controls.
- ADO.NET Architecture:
  - Connection Oriented: Understanding of Connection, Command, DataReader object.
  - o ConnectionLess: Understanding of DataAdapter, DataSet, DataTable and Dataview object.
- Working with XML – Overview of XML Classes and using XML with datasets.

Unit - IV: Overview of SOA Hours:04

- Overview of Service Oriented
  - Architecture o Service Provider
  - Service Consumer
  - Service

    - Service Description
    - SOAP
    - UDDI
- Building Web Service in ASP.NET
    - Deploying, Publishing and Consuming Web service.

Unit - V: Reporting Hours:05

Introduction to Crystal Report, Crystal Reports Architecture, ReportViewer Control, Object Model, Understanding Reporting Control

Unit - VI: Advanced ASP.NET Hours:05

- Introduction to ASP.NET AJAX.
- ASP.NET AJAX Control Toolkit Extender and controls.

Adjusting the Web Content, Validating Controls, Working with Menu, list and pop-ups. Introduction to MVC Architecture in ASP.NET,Developing web application in MVC Architecture.

Core Books:

1. Stephon Walther: ASP.Net Unleashed, BPB publication.
2. Kogent Solutions Inc.: ASP.Net 4.5 Black book, Dreamtech press, 2009.
3. Mridila Parihar, Essam Ahmed: ASP .Net Bible, Wiley, 2004.

Reference Books:

1. Mesbah Ahmed, Chris Garrett, Jeremy Faircloth, Chris Payne: ASP.Net Programming. Developer's Guide, Dreamtech, 1st Edition 2002.
2. A. Russell Jones, Mike Gunderloy: .Net Programming 10-Minute Solutions, BPB Publications.
3. Greg Buczek: ASP.Net Developer's Guide, Tata McGraw Hill 4th Edition, 2005.
4. Greg Buczek: ASP.Net Tips & Techniques, Tata McGraw Hill Edition, 2002.
5. Bolton, Justin Langford, Glenn Berry, Gavin Payne, Amit Banerjee, Rob Farley: Professional SQL Server 2012 internals and trouble shooting, Wiley India publication,October,2012.

Web References:

1. http://msdn.microsoft.com/en-us/aa336522.aspx [For Unit 1 and to download software]
2. http://www.asp.net/ [For Unit1 & 2 and to download software]
3. http://www.aspfree.com/ [Forum for discussion on ASP.NET]
4. http://www.devx.com/dotnet [To read latest published articles/news]
5. myweb.sabanciuniv.edu/gulsend/files/2010/03/intro.ppt [Lecture Note for Unit1 and 2]
6. www.cs.odu.edu/~mukka/cs795sum08/Lecturenotes/Day3/ado.ppt [Lecture Note for Unit 3]
7. http://www.codeproject.com/Articles/142064/Step-by-Step-Creation-of-Crystal-Reportusing-its [Lecture Note for Unit 5]

Course Outcomes: Upon successful completion of the course, the students will :

C01 : Understanding of .Net Framework

C02 : Students will be able to develop, test and deploy dynamic web applications

C03 : Understanding of database connectivity and creation of application

C04 : Students will be familiar with the concepts of SOA

C05 : Students will be familiar with the reporting tools

C06 : Students will be familiar with the concepts of AJAX and other advance .net technology

Course Outcomes Mapping:

Unit No.	**Unit Name**	**Course Outcomes**					
		C01	**C02**	**C03**	**C04**	**C05**	**C06**
1	Introduction to .NET framework and ASP.NET	✓					
2	Working with ASP.NET Applications		✓				
3	Database Connectivity through ADO.NET			✓			
4	Overview of SOA				✓		
5	Reporting					✓	
6	Advanced ASP.NET						✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	1	2	3	1	1	1	1	1	1
CO2	3	2	3	1	1	1	2	1	1
CO3	3	2	3	1	1	1	2	1	1
CO4	2	2	3	1	1	2	2	1	1
CO5	2	1	3	2	1	2	1	1	1
CO6	2	2	3	1	1	2	1	1	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA845: Advanced Database Technologies

(200 Marks)

Contact Hours: 06

Pre-requisite: Database Technologies

Methodology & Pedagogy: During theory sessions detailed understanding of query and transaction process mechanism, Database backup, recovery and security mechanism will be given. Students will also be taught how to write stored Procedures and how to trigger these procedures using specific procedural language.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	Practical
1	Introduction to PL\SQL	04	36
2	Transaction Management	08	
3	Distributed Transaction Management	07	
4	Object Relational Databases	07	
5	Data Warehousing, Data Mining & databases on the web	04	
6	Introduction to Big data and Hadoop	06	

Total Hours (Theory): 36

Total Hours(Lab):36

Total: 72

Detailed Syllabus:

Unit – I: Introduction to PL\SQL   Hours: 04

Introduction to PL/SQL,Cursor,Stored Procedure,Trigger and user defined Functions.

Unit – II: Transaction Management Hours: 08

Introduction to transaction, Recovery Techniques-Check pointing, Recovery from system crash, Media recovery, other approaches and interaction with concurrency control.

Unit – III: Distributed Transaction Management Hours: 07

Distributed transaction introduction, issue, distributed concurrency control, Distributed deadlock and deadlock detection, 2PC and recovery from failure, Presumed abort protocol.

Unit – IV: Object Relational Databases Hours: 04

Active Database, Temporal Database, Spatial databases, Mobile databases, Geographic information Database.

Unit – V: Data Warehousing, Data Mining & databases on the web Hours: 06

Introduction to data warehousing, components, characteristics, advantages and limitations of data warehouses, Introduction to data mining, goals of data mining, data mining techniques, data mining tools and applications, Overview of XML, Structure of XML data, Storage of XML data, XML applications.

Unit – VI: Introduction to Big data and Hadoop: 06

Types of Digital Data, Introduction to Big Data, Big Data Analytics, History of Hadoop, Apache Hadoop

Core Books:

1. Raghu Ramakrishnan,Johannes Gehrke:Database Management Systems,McGraw Hill Publication.
2. Ramez Elmasri,Shamkant B.Navathe:Fundamentals of Database Systems, 5th Edition, Pearson Publication.
3. S.K.Singh: Databse Systems,Concepts,Design and Applications, 1st Edition,Pearson Publication.
4. Tom White “Hadoop: The Definitive Guide:” Third Edit on, O’reily Media, 2012.
5. Seema Acharya, Subhasini Chellappan, "Big Data Analytics" Wiley 2015.

Reference Books:

1. Abraham Silberschatz,Henry F.Koeth,S.Sudarshan:Database System Concepts, 6th edition,McGraw Hill Publication.

Web References:

2. http://people.cs.aau.dk/~torp/Oracle/Introduction_to_plsql.pdf [Introduction to PL/SQL]
3. https://cs.nyu.edu/courses/fall09/G22.2434-001/index.html [Advanced Databse System]

4. http://infolab.usc.edu/csci585/Spring2010/den_ar/ordb.pdf [Object Relational Database]

Course Outcomes: Upon successful completion of the course, the students will :

C01 : Be familiar with PL\SQL programming.

C02 : Be familiar with Transaction management.

C03 : Be familiar with Distributed database and Distributed Transaction management.

C04 : Be familiar with various Object Relational databases.

C05: Be familiar with Advance Data warehousing, Data mining Techniques. Ad

Course Outcomes Mapping :

Unit No	Unit Name	C01	C02	C03	C04	C05
1	Introduction to PL\SQL	✓				
2	Transaction Management		✓	✓		
3	Distributed Transaction Management		✓	✓		
4	Object Relational Databases				✓	
5	Data Warehousing, Data Mining & databases on the web					✓
6	Introduction to Big data and Hadoop					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	3	2	2	-	-	2	-	2
CO2	2	-	2	2	2	-	-	2	-
CO3	2	2	3	2	2	2	2	-	2
CO4	3	3	3	-	2	2	-	-	-
CO5	-	-	3	3	2	3	2	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA846: Software Engineering Paradigms

(100 Marks)

Contact Hours: 03

Pre-requisite: Basic knowledge of System Analysis & Design

Methodology & Pedagogy: During theory lectures the emphasis will be given on the basics of software engineering concepts and UML. Some of the Agile software development concepts will be introduced during lectures. As part of modeling with UML students will be given exposure to model real-life software projects.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours Theory
1	Introduction to software engineering and process models	06
2	Unified Modelling Process & introduction to UML	07
3	Basics of Structural Modeling with UML	06
4	Basics of Behavioral Modeling with UML	07
5	Agile software development & Extreme Programming	06
6	Software Maintenance and Risk Management	05

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit – I Introduction to software engineering and process models Hours: 06

Defining software and software engineering, the software process, generic Process Models: the waterfall model, Incremental software process models, evolutionary process models

Unit – II Unified Modelling Process & introduction to UML Hours: 07

Unified process: The dynamic structure - Inception, Elaboration, Construction, and Transition phases; The static structure – different workflows, roles and artefacts. Examples

Overview of UML, Three steps of understanding the UML

Unit –III: Basics of Structural Modeling with UML Hours: 06

Modeling simple dependencies, modeling single inheritance Modeling structural relationships Structural Diagrams: Class Diagram, Examples

Unit –IV: Basics of Behavioral Modeling with UML Hours: 06

Behavioral Diagrams: Use Case Diagram, Interactions Diagrams: Sequence and collaboration diagram, Activity Diagram, Examples

Unit – V: Agile software development & Extreme Programming Hours: 06

Introduction to Agile Software Development, Characteristics of Agile Process, Agile methods , Principles of Agile methods, Problems with Agile methods, Extreme Programming, The Four Core Values of XP

Unit – VI: Software Maintenance and Risk Management Hours: 05

Software risks, risk identification, risk projection, risk mitigation, monitoring and management, software maintenance, software reengineering, reverse engineering

Core Books:

1. Grady Booch, James Rambaugh, Ivar Jacobson: The Unified Modeling Language User
2. Roger S. Pressman: Software engineering – A Practitioner's Approach, 7th Edition, ISBN: 978-007-126782-3, McGraw-Hill Publication.

Reference Books:

1. Ian Sommerville: Software Engineering, 8th Edition, ISBN: 978-81-317-2461-3, Pearson Education.
2. Guide, Addison Wesley. Jacobson, Booch, Rumbaugh, The Unified Software Development Process. Pearson Education, 1999.

Web References:

1. https://en.wikibooks.org/wiki/Introduction_to_Software_Engineering/Process/Methodology
2. [Introduction to Software Engineering/Process/Methodology]
3. https://www.uml-diagrams.org/ [UMLUnits]
4. https://nptel.ac.in/courses/106101061/26 [Agile Software Development and Extreme Programming]
5. https://nptel.ac.in/courses/Webcoursecontents/IIT%20Kharagpur/Soft%20Engg/pdf/m12L31.pdf [Risk Management]

Course Outcomes: Upon successful completion of the course, the students will :

C01 : Understanding Software Engineering Process Models

C02 : Start learning what is UML and how it is useful in system modeling

C03 : Be able to model using structural diagrams & behavioral diagrams

C04 : Understanding of the Agile software development concepts

C05 : Understanding the Risks associated with Software Development Process and managing them

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	**C02**	**C03**	**C04**	**C05**
1	Introduction to software engineering and process models	✓				
2	Unified Modelling Process & introduction to UML		✓			
3	Basics of Structural Modeling with UML			✓		
4	Basics of Behavioral Modeling with UML			✓		
5	Agile software development & Extreme Programming				✓	
6	Software Maintenance and Risk Management					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	1	-	-	-	1	2	-	-	-
CO2	2	2	2	3	2	2	-	2	-
CO3	2	2	2	3	2	2	-	2	-
CO4	2	2	3	-	2	2	2	2	3
CO5	2	2	-	-	1	2	2	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA847 : Cloud Computing (100 Marks)

Contact Hours: 03

Pre-requisite: Operating System Concepts and Network Technology.

Methodology & Pedagogy: During theory lectures the emphasis will be given on the fundamentals of cloud computing. Students will be introduced basic types, architecture, service providers, mechanism, security issues and some hidden aspects of cloud computing. Students will give practical exposure in form of case study and by showing cloud infrastructure of university.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours
		Theory
1	Evolution of Cloud Computing	04
2	Understanding Cloud Computing and basic types	07
3	Fundamentals of Cloud Architecture and Service Providers	07
4	Cloud Computing Mechanisms	07
5	Cloud Computing Security and Business Use	06
6	Hidden Aspects of Cloud Computing	05

Total Hours (Theory): 36
Total: 36

Detailed Syllabus:

Unit – I: Evolution of Cloud Computing Hours: 04
Introduction of Cloud Computing, Growth of technology, Paradigm Shift in Computing, distributed nature of service Provisioning, Support entrepreneurship using Cloud Computing.

Unit – II: Understanding Cloud Computing and basic types Hours: 07
Advantages and drawbacks of Cloud Computing, Essential component for Cloud contract, Major outage of Cloud Computing and Enhancers for Cloud Computing.
Introduction to SaaS, PaaS, IaaS. Introduction to Public Cloud, Private Cloud, Hybrid Cloud and Community Cloud, Storage Services for Cloud Computing

Unit – III: Fundamentals of Cloud Architecture and Service Providers Hours: 07
Workload Distribution Architecture, Resource Pooling Architecture, Dynamic Scalability Architecture, Elastic Resource Capacity Architecture, Service Load Balancing Architecture, Cloud Bursting Architecture, Elastic Disk Provisioning Architecture, Redundant Storage Architecture.
Introduction to major Cloud Service Provider: Amazon Web Services, Google Cloud.

Microsoft Windows Azure and Office 365, Hp Cloud, RackSpace, CSC Corp, Verizon Terrimark, DropBox.

Unit – IV: Cloud Computing Mechanisms Hours: 07

Introduction to Cloud Infrastructure Mechanisms: Logical Network Perimeter, Virtual Server, Cloud Storage Device, Cloud Storage Levels, Network Storage Interfaces, Object Storage Interfaces, Database Storage Interfaces, Relational Data Storage, on-Relational Data Storage, Cloud Usage Monitor, Monitoring Agent, Resource Agent, Polling Agent, Resource Replication. Introduction to Cloud Management Mechanisms: Remote Administration System, Resource Management System, SLA Management System, Billing Management System.

Unit – V: Cloud Computing Security and Business Use Hours: 06

Introduction to Encryption, Symmetric Encryption, Asymmetric Encryption, Hashing, Digital Signature, Public Key Infrastructure (PKI), Identity and Access Management (IAM), Single SignOn (SSO), Cloud-Based Security Groups. Overview of Compliance and Certification, Access Control, Organizational Control.

Benefits of Business using Cloud Computing, Risk of Cloud Computing, Cost factor in Cloud Computing.

Unit – VI: Hidden Aspects of Cloud Computing Hours: 05

Introduction to Hidden Aspects of Cloud Computing, Service level Agreement, Sharing Log Data, Service Uptime Guarantee.

Core Books:

1. S. Srinivasan: Cloud Computing Basics, Springer,2014.
2. Thomas Erl, Zaigham Mahmood and Ricardo Puttini: Cloud Computing Concepts, Technology & Architecture, PHI, 2013.

Reference Books:

1. Derrick Rountree, Ileana Castrillo : The Basics of Cloud Computing, Syngress, 2013.
2. Rajkumar Buyya, James Broberg, Andrzej M. Goscinsk: Cloud Computing- Principles and Paradigms, John Wiley &Sons, 2011.

Web References:

1. http://whatisCloud.com/basic_concepts_and_terminology/Cloud [For basic terminology of Cloud Computing]
2. http://www.tutorialspoint.com/Cloud_Computing/ [For cloud computing lecture notes]
3. http://www.intel.in/content/dam/www/public/us/en/documents/guides/cloud-computingvirtualization-building-private-iaas-guide.pdf [For cloud computing virtualization]
4. www.cs.purdue.edu/.../Anya-Kim-Bhargava-MCCWorkshop.ppt [Security issues PPTs]

Course Outcomes: Upon successful completion of the course:

C01 : Students will learn basics of cloud computing.

C02 : The students will be familiar with various cloud architectures and services.

C03 : They will get the knowledge of various network mechanics.

C04 : Students will get various business aspects of cloud computing with security aspects.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes			
		C01	**C02**	**C03**	**C04**
1	Evolution of Cloud Computing	✓			
2	Understanding Cloud Computing and basic types		✓		
3	Fundamentals of Cloud Architecture and Service Providers		✓		
4	Cloud Computing Mechanisms			✓	
5	Cloud Computing Security and Business Use				✓
6	Hidden Aspects of Cloud Computing				✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	-	-	1	-	1	1	1	-	-
CO2	3	2	3	1	1	1	1	-	-
CO3	1	-	2	2	-	-	1	-	-
CO4	1	1	1	2	1	1	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA848 : Cyber Security and Computer Forensics

(100 Marks)

Contact Hours: 03

Pre-requisite: Basic knowledge of fundamental Networking.

Methodology & Pedagogy: During theory lectures the emphasis will be given on the basics of cyber crime, tools and techniques used in cyber crime, devices used to perform cyber crime, detection and prevention of cyber crimes with wired and wireless devices. The laws prevailing for cyber crimes are also discussed. Examples and Mini-Cases and online scams will be discussed to enhance understanding. The teacher will also introduce the students to the cyber forensics concepts and tools.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours Theory
1	Introduction to Cyber Crime	07
2	Tools and Techniques Used in Cyber Crime	06
3	Wireless Device and Cyber Crime	06
4	Cyber Security, Corporate Security and Legal Aspects	07
5	Introduction to Computer Forensics	05
6	Forensics of Wireless Devices	05

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit – I: Introduction to Cyber Crime   Hours: 07

Introduction to cyber crimes , Distinction between cyber crime and conventional crimes, Reasons for commission of cyber crime, Common Cyber Threats
Classification Of Cyber Crimes: Cyber crime against Individual, Cyber crime Against Property,
Cyber crime Against Organization, Cyber crime Against Society
Classifications of Cybercrimes: E-Mail Spoofing,
Spamming, Cyber defamation, Internet Time Theft, Newsgroup Spam/Crimes from Usenet Newsgroup, Industrial Spying/Industrial Espionage, Hacking, Online Frauds, Pornographic Offenses , Software Piracy, Password Sniffing, Credit Card Frauds and Identity Theft.

Unit – II: Tools and Techniques Used in Cyber Crime   Hours: 06

Cyber offenses: Cyber criminals, organized cyber crimes, Types of attacks, Botnet.
Methods Used for Cybercrime: Proxy Servers and Anonymizers, Password Cracking, Phishing, Key loggers, Spywares Trojan
Attacks on Wireless Networks: Traditional Techniques of Attacks on Wireless Networks, Theft of Internet Hours and Wi-Fi-based Frauds and Misuses.

Unit – III: Wireless Device and Cyber Crime Hours: 06

Introduction: Proliferation of Mobile and Wireless Devices, Mobile Phone Theft, Mobile Viruses, Mishing, Vishing, Smishing, Hacking Bluetooth
Credit Card Frauds in Wireless Device: Types and Techniques of Credit Card Frauds, Security Challenges Posed by Mobile Devices, Protecting Data on Lost Devices.

Unit – IV: Cyber Security, Corporate Security and Legal Aspects Hours: 07

Cyber Security (IT security), Security principles, Security triad: Confidential, Integrity, Availability, Security Policy**,** Security Service Life Cycle
Aspects of Organizational Security- Information Security, Information Security's Overview and Services, Physical security, E-commerce Security, Legal security, Email security, Goals of Security.
Provisions in Indian Laws for dealing with Cyber Crimes

Unit – V: Introduction to Computer Forensics Hours: 05

The Need for Computer Forensics, Forensics Analysis of E-Mail : RFC282, Network Forensics, Computer Forensics and Steganography, Forensics and Social Networking Sites.
Challenges in Computer Forensics, Special Tools and Techniques, Forensics Auditing and Antiforensics.

Unit – VI: Forensics of Wireless Devices Hours: 05

Digital forensics and wireless devices, Toolkits for Hand-Held Device Forensics: EnCase, Device Seizure and PDA Seizure, Palm DD, Forensics Card Reader, Cell Seizure, MOBILedit!, ForensicSIM, Organizational Guidelines on Cell Phone Forensics: Hand-Held Forensics as the Specialty Domain in Crime Context.

Core Books:

1. Nina Godbole, Sunit Belapur: "Cyber Security Understanding Cyber Crimes, Computer Forensics and Legal Perspectives", Wiley India Publications, April 2011.
2. Robert Jones: "Internet Forensics: Using Digital Evidence to Solve Computer Crime", O'Reilly Media, October, 2005.
3. Harish Chander " Cyber Laws and IT Protection", PHI Learning, 2012.

Reference Books:

1. Chad Steel: "Windows Forensics: The field guide for conducting corporate computer investigations", Wiley India Publications, December, 2006.
2. Eoghan Casey: "Digital Evidence and Computer Crime", 3rd Edition, Academic Press, 2011.

Web References:

1. http://www.lawyersclubindia.com/articles/Classification-Of-CyberCrimes-
2. 1484.asp#.VWBGbdKqqko [Classification Of Cyber Crimes]
3. http://www.cyberlawclinic.org/cybercrime.htm [Details regarding Cyber crime, laws, case studies, etc.]
4. https://www.us-cert.gov/sites/default/files/publications/forensics.pdf [Legal Aspects and Resources of Computer Forensics]
5. https://leocybersecurity.com/wp-content/uploads/2017/11/16-Digital-Forensic-Tools.pdf [Digital Forensics Tools]

Course Outcomes: Upon successful completion of the course, the students will :

C01 : Distinguish between different types of cyber crimes

C02 : Be familiar with tools and techniques employed for cyber crime

C03 : Be familiar with different cyber security techniques for individual corporates and legal provision for cyber crime.

C04 : Be familiar with the concepts of computer forensics and various related tools.

C05 : Be able to perform independent research in cyber security and forensics.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	**C02**	**C03**	**C04**	**C05**
1	Introduction to Cyber Crime	✓				✓
2	Tools and Techniques Used in Cyber Crime		✓			✓
3	Wireless Device and Cyber Crime	✓				✓
4	Cyber Security, Corporate Security and Legal Aspects			✓		✓
5	Introduction to Computer Forensics				✓	✓
6	Forensics of Wireless Devices				✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	2	3	2	-	3	2	-	3	-
CO2	3	3	-	2	2	-	2	-	3
CO3	-	2	3	3	-	3	-	1	2
CO4	2	2	-	2	3	-	3	-	1
CO5	-	-	2	3	-	2	-	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA849 : Internetworking with TCP/IP Protocol Suite (100 Marks)

Contact Hours: 03

Prerequisites: Fundamentals of Computer Networks.

Methodology & Pedagogy: During theory sessions the students shall be introduces to various internet and intranet technologies and network services. As a case study an institute level network design, devices, components and security mechanism will be demonstrated. During practical sessions students will be trained to develop network based applications using available technologies.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours Theory
1	Addressing Techniques	06
2	Internet Protocol (TCP/IP)	06
3	Routing	07
4	UDP and Mobile IP	06
5	Cellular Systems	05
6	Internet Security	06

Total Hours (Theory): 36

Total: 36

Unit 1 Addressing Techniques Hours:06

Classful Internet Addresses, Mapping Internet Addresses (ARP), Determining An Internet Address (RARP)

Unit 2 Internet Protocol (TCP/IP) Hours:06

Internet Protocol: Connectionless Datagram Delivery, Internet Protocol: Routing IP Datagrams, Internet Protocol: Error and control Messages, ICMP

Unit 3 Routing Hours:07

Congestion, Router Discovery and solicitation, Classless and Subnet Address Extensions, Transparent routers, Proxy ARP and Subnet addressing, CIDR, Subnet routing. Static and Dynamic routing, Difference between static and dynamic routing. Introduction to dynamic routing algorithms. Distance vector and link state routing algoritms.

Unit 4 UDP and Mobile IP Hours:06

User Datagram Protocol, Reliable Stream Transport Service, Sliding Window, TCP, Karn's Algo, Congestion, RED Silly window syndrome, starting and closing TCP connection, Mobile IP

Unit 5 Cellular Systems Hours:05

Cellular system overview - Cellular system organization, Frequency Reuse, Increasing Capacity, Operation of cellular system, steps in an Mobile switching center(MSC) controlled call between mobile users, Mobile Radio Propagation effect, Additional Function in MSC Controlled Call, Handoff Performance metrics, Handoff Strategies Used to Determine Instant of Handoff, Power control, Traffic Engineering.

Unit 6 Internet Security Hours: 06

Internet Security, Firewalls. Introduction to cryptography and SSL.
Applications: World Wide Web, Electronic mails, FTP, Remote login

Core Book:

1. Duglas E Comer: Internetworking with TCP/IP Volume – I, Fourth Edition, PHI.
2. Andrew S. Tanenbaum, David Wetherall: Computer Networks, PHI
3. Ajit Pal: Data Communications and Computer Networks, PHI

Reference Book :

1. Behrouz A. Forouzan: TCP / IP Protocol Suite, Third Edition
2. Natalia Olifer, Victor Olifer: Computer Networks: Principles, Technologies and Protocols for Network Design, Willy
3. Debra Littlejohn Shinder: Computer Networking Essentials: An essential guide to understanding networking theory, implementation and interoperability, Cisco Press.

Web References:

1. http://www.firewall.cx [Basics of Network Protocols]
2. http://nptel.ac.in [Video Lectures]
3. https://www.cs.purdue.edu/homes/comer/netbooks.html [Teaching Material]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Better understanding of classful and classless addressing scheme.

C02 : Be familiar with TCP/IP reference model.

C03 : To understand different routing protocols.

C04 : To familiar themselves with mobility in communication.

C05 : To familiar themselves with cellular systems communication.

C06: To familiar themselves with network security.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes					
		C01	**C02**	**C03**	**C04**	**C05**	**C06**
1	Addressing Techniques	✓					
2	Internet Protocol (TCP/IP)		✓				
3	Routing			✓			
4	UDP and Mobile IP				✓		
5	Cellular Systems					✓	
6	Internet Security						✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	2	2	2	2	1	-	1	-	-
CO2	2	2	2	2	1	2	1	-	-
CO3	2	2	2	2	1	-	1	-	-
CO4	2	2	2	2	1	-	1	-	-
CO5	2	2	2	2	1	-	1	-	-
CO6	2	2	2	2	1	-	3	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA850: Advanced Operating Systems
(100 Marks)

Contact Hours: 03

Pre-requisite: Operating Systems concepts

Methodology & Pedagogy: During theory lectures the emphasis will be given on the advanced concepts of Operating Systems. Students will be introduced jargons of distributed and mobile based operating systems. During theory lectures concepts of shared memory, remote procedure call, synchronization, process management, resource management and distributed file systems will be discussed. Students will also give overview of mobile based operating systems.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours
		Theory
1	Overview of Distributed Systems	04
2	Distributed Shared Memory and Remote Procedure Calls	08
3	Synchronization in Distributed Operating Systems	06
4	Process and Resource Management in Distributed Operating Systems	08
5	Distributed File Systems	05
6	Mobile and Real time Operating Systems	05

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit – I: Overview of Distributed Systems Hours: 04
What is Distributed operating system, issues in designing distributed operating system – transparency, Reliability, Flexibility, performance, scalability, heterogeneity, security, Emulation of existing operating system, Introduction to distributed computing environment.

Unit – II: Distributed Shared Memory and Remote Procedure Calls Hours: 08
Introduction to RPC, RPC Model, Implementing RPC Mechanism, Stub Generation, RPC Message, Introduction to Distributed Shared Memory, General Architecture of DSM Systems, Design and implementation issues of DSM, Granularity, Structure of Shared memory space.

Unit – III: Synchronization in Distributed Operating Systems Hours: 06

Clock Synchronization – Implementation, drifting of clocks, Clock synchronization issues, Clock Synchronization algorithms, Mutual Exclusion – Centralized approach, Distributed Approach, Token Passing Approach.

Unit – IV: Process and Resource Management in Distributed Operating Systems Hours: 08
Introduction to resource management in distributed operating system, Desirable features of good global scheduling algorithm, Task Assignment approach, Load balancing Approach, Load Sharing Approach, Process migration – desirable feature of Process migration, process migration mechanisms, Process migration in heterogeneous systems, Advantages of process migration.

Unit – V: Distributed File Systems Hours: 05
What is Distributed File system – Remote information sharing, User mobility, Availability, Diskless workstation. Types of services in distributed file system – Storage service, True file Service, Name Service. Desirable features of Distributed file system, File Models, File accessing models.

Unit – VI: Mobile and Real Time Operating Systems Hours: 05
Introduction to Mobile Phone Systems, Memory, Scheduling and Concurrency management in iOS and Android OS ,Overview of Real time OS.

Core Books:

1. Pradip K. Sinha: Distributed Operating Systems Concepts and Design, Eastern Economy Edition, PHI, 2007.
2. Michael J.Jipping : Smartphone Operating System Concepts with Symbian OS, Wiley, 2007.

Reference Books:

1. Andrew S. Tanenbaum: Distributed Operating Systems, 5th Edition, Pearson, 2008.
2. Doreen L. Galli: Distributed Operating Systems Concepts and Practice, Prentice Hall, 2000.
3. Andrew S. Tanenbaum, Herbet Bos: Modern Operating Systems, 4th Edition, Pearson Education, 2014.

Web References:

1. www.cs.columbia.edu/smb/classes/s06-4118/l26.pdf [Overview of Distributed OS]
2. www.cs.uah.edu/~weisskop/Notes690/A5_DistSysCh1.ppt [PPTs of Distributed OS]
3. https://www.elprocus.com/real-time-operating-system-rtos-and-how-it-works/ [Realtime OS and its working]
4. https://www.tutorialspoint.com/computer_fundamentals/mobile_operating_systems.asp [Mobile OS tutorials]

Course Outcomes: Upon successful completion of the course, the students will :

C01 : Be familiar with Distributed Operating Systems and their designing issues

C02 : Be familiar with the functionalities of Distributed Operating Systems

C03 : Be Familiar with Mobile and Real Time Operating systems with their functionalities

C04 : Be able to design the functionality of advanced operating systems

C05 : Be able to perform independent research in distributed, mobile and real time Operating Systems.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	**C02**	**C03**	**C04**	**C05**
1	Overview of Distributed Systems	✓			✓	✓
2	Distributed Shared Memory and Remote Procedure Calls	✓	✓		✓	✓
3	Synchronization in Distributed Operating Systems	✓	✓		✓	✓
4	Process and Resource Management in Distributed Operating Systems	✓	✓		✓	✓
5	Distributed File Systems	✓	✓		✓	✓
6	Mobile and Real time Operating Systems			✓	✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	2	2	2	2	1	-	1	-	-
CO2	2	2	2	2	1	2	1	-	-
CO3	2	2	2	2	1	-	1	-	-
CO4	2	2	2	2	1	-	1	-	-
CO5	2	2	2	2	1	-	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS704 C: ACADEMIC SPEAKING (Sem-I)

I. Credits and Schemes:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
1	HS704 A/B/C/D/E/F/G/H	Academic Speaking	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Hours
1	**Foundations of Advance Communication** • *Meaning and Definition of Advance Communication* • *Advance Communication in Digital, Social, Mobile World* • *Strategies for Advance Communication* • Meaning and Concept of Academic Language • High Frequency Academic Vocabulary	**04**
2	**Art of Conversation** • *Describing people, places and things* • *Expressing opinions* • *Making suggesting* • *Persuading someone* • *Interpreting and Summarizing*	**06**

<table>
<tr><td>3</td><td>Science of Power Speaking
Phonemes
Word Stress
Pronunciation
Intonation
Pause
Register
Fluency
Prosody
Lexical Range</td><td>06</td></tr>
<tr><td>4</td><td>Academic Speaking Application – Part I
Art of Oratory
Formal Presentation
Speech Analysis – Decoding Best Speeches</td><td>08</td></tr>
<tr><td>5</td><td>Academic Speaking Application – Part II
Job Interview
Group Discussion
Meeting</td><td>06</td></tr>
<tr><td colspan="2" align="right">Total</td><td>30</td></tr>
</table>

III. **Instruction Methods and Pedagogy**

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like case studies, group work, independent and collaborative research, presentations etc.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

<table>
<tr><th>Sl. No.</th><th>Component</th><th>Number</th><th>Marks per incidence</th><th>Total Marks</th></tr>
<tr><td>1</td><td>I-Talk</td><td>1</td><td>10</td><td rowspan="3">25</td></tr>
<tr><td>2</td><td>Situational Speaking</td><td>1</td><td>05</td></tr>
<tr><td>3</td><td>Case Study - Speech Analysis</td><td>2</td><td>10</td></tr>
<tr><td>4</td><td>Attendance and Class Participation</td><td colspan="2">-</td><td>05</td></tr>
<tr><td colspan="4" align="right">Total</td><td>30</td></tr>
</table>

External Evaluation

The University Practical Examination will be for 70 marks and will test the advance communication skills and academic speaking.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
			Total	**70**

Course Outcome (COs):

After completion of the course the student would :

CO1 understand and demonstrate advance communication skills and academic speaking.
CO2 demonstrate linguistic competence
CO3 demonstrate performing ability at group discussion and personal interview.
CO4 demonstrate the formal presentation skills.
CO5 demonstrate ability to communicate in diverse situations

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	3	-	-	-	-
CO3	-	-	-	-	3	2	-	-	-
CO4	-	-	-	-	3	-	-	-	-
CO5	-	-	-	-	3	-	-	-	2

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

V. Reference Books

- Headway Academic Skills - Level 1: Listening, Speaking and Study Skills Student's Book Paperback

VI. Reading

- **Unit 1:** Business communication Today (Thirteenth Edition) by Courtland L. Bovee, John V. Thill and Roshan Lal Raina
- **Unit 2:** Effective Speaking Skills by Terry O' Brien
- **Unit 2:** Speak Better Write Better by Norman Lewis
- **Unit 2:** Well Spoken: Teaching Speaking to All Students by Erik Palmer
- **Unit 3:** Let Us Hear Them Speak : Developing Speaking – Listening Skills in English by Jayshree Mohanraj (Publisher – Sage Publication)
- **Unit 4:** The craft of scientific presentations: Critical steps to succeed and critical errors to avoid. New York: Springer by Michael Alley
- **Unit 4:** Presentation Skills in English by Bob Dignen (Publisher: Orient Black Swan)

OBJECTIVES, TEACHING SCHEME & DETAILED SYLLABUS

FOR

MCA PROGRAMME
(2nd SEMESTER)

EFFECTIVE FROM
ACADEMIC YEAR 2019-20

Teaching and Examination Scheme (MCA Programme)
Effective from Year 2019-20
Semester-II

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA851	Web Application Development using Open Source Technology	3	3	6	6	10	20	70	15	15	70	200
CA852	Software Quality Assurance	3	3	6	6	10	20	70	15	15	70	200
CA853	Responsive Web Designing Using Framework	-	2	2	2	-	-	-	15	15	70	100
CA854-CA855	**Elective II**	3	3	6	6	10	20	70	15	15	70	200
CA856-CA858	**Elective-III**	3	3	6	6	10	20	70	15	15	70	200
HS705 C	Academic Writing	-	2	2	2	-	-	-	30		70	100
	University Elective-II **	-	2	2	2	-	-	-	30		70	100
		12	18	30	30	400			700			1100

** Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **Mobile Application Development** course for others.

Elective-II

1. CA854 Mobile Application Technology – Android Platform
2. CA855 Mobile Application Development-iOS Platform

Elective-III

1. CA856 HTTP Web Service for Enterprise Application
2. CA857 Framework and Applications
3. CA858 Multi Paradigm Scripting using Python

University Elective-II

No	Course Code	Name	Department/Faculty
1	EE782.01	Energy Audit and Management	Engineering
2	CE771.01	Project Management	Engineering
3	PT796.01	Fitness & Nutrition	Physiotherapy
4	NR752.01	Epidemiology and Community Health	Nursing
5	OC733.01	Introduction to Polymer Science	Applied Science
6	MB651	Software based Statistical Analysis	Management
7	PH892	Intellectual Property Rights	Pharmacy
8	MA772.01	Design of Experiments	Mathematics
9	PD262	Astrophysics, Space and Cosmos-II	Applied Science

CA851 : Web Application Development using Open Source Technology

(200 Marks)

Contact Hours: 06

Pre-requisite: Basic understanding of HTML and MySQL.

Methodology & Pedagogy: In theory sessions, the emphasis will be given on introduction to open source technology, the structure and syntax of PHP, database connectivity using SQL and No SQL databases, working with forms and user data, form and error handling, object oriented programming with PHP, XML, Parsing an XML Document and responsive we applications. During Practical sessions, students will implement the concepts, which are taught during theory sessions.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1	Introduction and Working with PHP	08	36
2	Web Techniques and Form Handling using PHP	06	
3	Accessing Relational Databases using PHP	06	
4	Accessing NoSQL Databases using PHP	06	
5	Working with XML and PHP	06	
6	Responsive Web Application Development using AJAX and PHP	04	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total: 72

Detailed Syllabus:

Unit – I: Introduction and Working with PHP Hours: 08

Overview of Open Source Software, Installation & Configuration of PHP, Introduction to PHP. PHP language Basics: Lexical Structure, Data types, Variables, Expressions and Operators, Control and Looping statements. Functions: Function Definition, Function Parameters, Returning Values. Strings: Usages and String Functions. Arrays: Types of Arrays and its Usages, Array functions. Objects: Declaring Class, Properties, Methods, Exception Handling, Examples.

Unit – II: Web Techniques and Form Handling using PHP Hours: 06

HTTP & HTTP2 Basics, Super Global Variables, Processing Forms, Setting Response Headers, State Management Techniques in PHP, Form validation using Regular Expressions.

Unit - III: Accessing Relational Databases using PHP Hours: 06

Using PHP to access Databases, Relational Databases and SQL, PHP Data Objects (PDO), MySQLi Object Interface.

Unit – IV: Accessing NoSQL Databases using PHP Hours: 06

Introduction to No SQL Databases, Using PHP to access No SQL Databases, MongoDB Database and its configuration, MongoDB Basics, Accessing MongoDB database using PHP and CRUD Operations.

Unit – V: Working with XML and PHP Hours: 06

XML: Introduction to XML, Generating XML, Parsing XML, Parsing XML with DOM, parsing with SimpleXML, Transforming XML with XSLT. Web Services: REST Clients, XML – RPC.

Unit VI: Responsive Web Application Development using AJAX and PHP Hours: 04

Introduction to AJAX, PHP and AJAX Example, AJAX Suggest and Autocomplete, AJAX Data Grid, Introduction to PHP Web Sockets.

Core Books:

1. Kevin Tatroe, Peter MacIntyre, Rasmus Lerdorf: Programming PHP - Creating Dynamic Web Pages, 3rd Edition, Kindle Edition, O'REILLY Publication.
2. David Sklar and Adam Trachtenberg: PHP CookBook - Solutions and Examples for PHP Programmers, 3rd Edition, O'REILLY Publication.
3. Christian Darie, Brinzarea Bogdan, Filip Chereches-Tosa, Mihai Bucicia: AJAX and PHP: Building Responsive Web Applications, Kindle Edition, Packt Publishibng.

Reference Books:

1. Matt Doyle: Beginning PHP 5.3, Wrox Publication, 2010.
2. Steve Francia: MongoDB and PHP, O'Reilly Media Publication.

Web References:

1. http://web-algarve.com/books/MySQL%20&%20PHP/PHP%20Cookbook,%203rd%20Edition.pdf [PHP basics and Examples]
2. https://www.w3schools.com/pHP/default.asp [For Basics of PHP]
3. https://www.tutorialspoint.com/php/index.htm [For State Management techniques]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Understand the basics of PHP and its working.

C02 : Be able to handle HTML Form and process user input using PHP.

C03 : Be able to develop interactive and dynamic web based application using PHP and MySQL, MongoDB

CO4 : Understand the concept of Interoperability and XML parsing

CO5: Be able to develop the responsive web applications

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	**C02**	**C03**	**C04**	**C05**
1	Introduction and Working with PHP	✓		✓		
2	Web Techniques and Form Handling using PHP		✓	✓		
3	Accessing Relational Databases using PHP		✓	✓		
4	Accessing NoSQL Databases using PHP		✓	✓		
5	Working with XML and PHP			✓	✓	
6	Responsive Web Application Development using AJAX and PHP			✓		✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	-	3	-	-	-	-	-	-
CO2	-	3	2	2	-	-	-	-	-
CO3	3	3	-	3	2	-	2	1	1
CO4	-	-	3	-	3	-	2	1	1
CO5	-	-	-	-	-	3	2	1	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA852 : Software Quality Assurance

(200 Marks)

Contact Hours: 06

Pre-requisite: Software Engineering.

Methodology & Pedagogy: The theory sessions will cover the software testing concepts and practices that support the production of quality software. The practical sessions will cover the application of testing techniques at various levels of testing using manual and automated testing tools.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	Practical
1	Introduction to Software Testing	05	36
2	Test Case Design Structural Techniques	05	
3	Test Case Design Behavioral Techniques	07	
4	Levels of Testing	07	
5	Software Testing Life Cycle	08	
6	Quality Assurance and Quality Control	04	

Total Hours (Theory): 36
Total Hours (Lab): 36
Total: 72

Detailed Syllabus:

Unit – I: Introduction to Software Testing Hours: 05

Importance of testing, Testing roles and responsibilities, software testing principles, concept of quality. Levels of testing, Software testing methodologies: White Box Testing, Black Box Testing, Grey Box Testing.
Design test case: Stateless and State oriented test cases.
Introduction to Test case design techniques, Static Techniques: Informal Reviews, Walkthroughs, Technical Reviews, Inspection. Introduction to Dynamic Techniques.

Unit – II: Test Case Design Structural Techniques Hours: 05

Overview of White Box, Control flow testing: Statement Coverage Testing, Branch Coverage Testing, Path Coverage Testing, Conditional Coverage Testing

Data flow testing: Data Flow Anomaly, Overview of Dynamic Data Flow Testing, Data Flow Graph, Data Flow Terms, Data Flow Testing Criteria.

Unit – III: Test Case Design Behavioral Techniques Hours: 07

Overview of Black Box: Equivalence Class Partition, Boundary Value Analysis, Pairwise Technique, Cause Effective Graph, Decision Table.
Experienced Based Techniques: Error guessing, Exploratory testing

Unit – IV: Levels of Testing Hours: 07

Introduction to functional and non functional testing
Functional Testing : Unit Testing, Integration Testing, System Testing, User Acceptance Testing. Sanity/Smoke Testing, Regression Test, Retest.
Non Functional Testing: Performance Testing, Scalability Testing. Compatibility Testing, Security Testing, Recovery Testing, Installation Testing.

Unit – V: Software Testing Life Cycle Hours: 08

Requirements Analysis/Design : Understand the requirements, Prepare Traceability Matrix Test Planning **:** Object, Scope of Testing, Schedule, Approach, Roles & Responsibilities , Assumptions, Risks & Mitigations, Entry & Exit Criteria, Test Automation, Deliverables. Test Cases Design**:** Write Test cases, Review Test cases, Test Cases Template, Types of Test Cases, Difference between Test Scenarios and Test Cases.
Test Environment setup: Understand the SRS, Hardware and software requirements, Test Data Test Execution: Execute test cases, Defect Tracking and Reporting: Types of Bugs, Identifying the Bugs, Bug/Defect Life Cycle, Reporting the Bugs, Severity and priority. Test Closure: Criteria for test closure, Test summary report
Test Metrics: Test Measurements, significance of Test Metrics, Metric Life Cycle, Types of Manual Test Metrics.

Unit – VI: Quality Assurance and Quality Control Hours: 04

Introduction to Quality Assurance and Quality Control.

Concept of Quality Assurance, Quality Control, Differences between QA & QC.

Core Books:

1. Sagar Naik, Piyu Tripathy: Software Testing and Quality Assurance, Theory and Practice, Wiley, 2008.
2. Paul C. Jorgensen : Software Testing: A Craftsman's Approach, 4th Edition by , CRC press, Taylor and Francsis Group,2014
3. Roger S Pressman: Software Engineering – A Practitioner's Approach, 7th Edition, McGRAW HILL International Edition, 2010.

Reference Books:

1. Mauro Pezze, Michael Young : Software testing and Analysis- Process, Principles and Techniques, Wiley India, 2012.

2. Boris Beizer: Software Testing Techniques: 2nd Edition, Van Nostrand Reinhold, 1990.
3. Daniel Galin: Software Quality Assurance, Pearson Education, ,2004.
4. Ron Patton: Software Testing, Pearson Education, 2001.

Web References:

1. https://www.softwaretestinghelp.com/quality-assurance-vs-quality-control
   [Introduction to Software Quality Assurance]
2. http://tryqa.com/what-is-software-testing [basic of Software Testing]
3. https://www.guru99.com/functional-testing.html [Functional Testing]
4. http://www.softwaretestinggenius.com/download/bgstpadmini.pdf [Software Testing Life Cycle]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Understand the role of testing in software development.

C02 : Apply the test case design techniques.

C03 : Acquire the various levels of testing.

C04 : Working knowledge of Software Testing Life Cycle.

C05 : Understand the importance and differences between Quality Assurance and Quality Control **.**

Course Outcomes Mapping:

Unit No.	**Unit Name**	**Course Outcomes**				
		C01	**C02**	**C03**	**C04**	**C05**
1	Introduction to Software Testing	✓				
2	Test Case Design Structural Techniques	✓	✓			
3	Test Case Design Behavioral Techniques	✓	✓			
4	Levels of Testing	✓		✓		
5	Software Testing Life Cycle	✓			✓	

6	Quality Assurance and Quality Control					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	3	3	3	3	3	3	3	3
CO2	3	2	2	3	3	2	3	3	3
CO3	3	3	3	2	2	3	2	3	3
CO4	3	2	3	3	3	3	3	3	3
CO5	3	3	2	3	3	3	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA853 : Responsive Web Designing Using Framework
(100 Marks)

Contact Hours: 02

Pre-requisite: Basic knowledge of HTML5, CSS3 and JavaScript.

Methodology & Pedagogy: During Practical sessions students will be required to design and develop responsive web sites using HTML5, CSS3, AngularJS and NodeJS.

Outline of the Course:

Week No.	Content
1	• Introduction to JavaScript & AngularJS, introduction to React JS and Vue JS. • Creating basic AngularJS application using its component, How AngularJS Integrates with HTML. • Working with AngularJS Expressions
2-3	• Data Binding in AngularJS. • Working with AngularJS Directives. • Working with the variables. • Creating a loop in AngularJS. Handling Events
4-6	• Working with Controller. • MVVM design Architecture with AngularJS. • Working with AngularJS Module. • Working with forms and its elements.
7-8	• Working with AngularJS HTML DOM • Working with Database using AngularJS.
9-10	• Fundamentals of Node JS, Installation of Node JS Basics of Node JS module. • Node JS Events. • Developing Model-View-Controller layer.
11-12	• Database connectivity with Node.js • Create database, create table, Insert, Select Form, where ,order by,delete,drop table, update, join, limit

Total Hours :24

Core Books:

1. Jeffry Houser : “Learn With: Angular 5, Bootstrap, and NodeJS”, Kindle Edition,

2. Shyam Seshadri Brad Green:"AngularJS – Up and Running, Brad Green", Second Edition, O'REILLY
3. Basarat Ali Syed: "Beginning Node.js", Apress Publication.

Reference Books:

1. Agus Kurniawan: "AngularJS Programming by Example 2017 Edition", Kindle Edition.
2. Adam Freeman: " Pro AngularJS 2017 Edition", Apress.
3. Krasimir Tsonev: "Node.js By Example", Packt Publishing

Web References:

1. http://www.w3schools.com/angular/default.asp [Tutorial link for AngularJS]
2. http://www.tutorialspoint.com/angularjs/ [Tutorial link for AngularJS]
3. https://www.tutorialspoint.com/angularjs/angularjs_tutorial.pdf [E-book for AngularJS]
4. http://www.tutorialsteacher.com/nodejs/nodejs-modules [Tutorial link for NodeJS]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Be able to design responsive web page for different devices. (desktops, tablets, and phones)

C02 : Be familiar with JavaScript Framework (AngularJS and NodeJS) and its applications.

C03 : Be able to create a single page application using AngularJS.

C04 : Create web application using the MVVM pattern with JavaScript Framework. Also, able to maintain the two way data binding between model and view.

C05 : Be able to create responsive web page with database using AngularJS and NodeJS.

Course Outcomes Mapping:

Lab	Content	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to JavaScript Framework.	✓	✓			
2-3	Basics of AngularJS .		✓	✓		
4-6	Working With MVVM Architecture.			✓	✓	
7-8	Working with AngularJS HTML DOM & Database.	✓		✓	✓	✓
9-10	Basic Fundamental of NodeJS		✓		✓	
11-12	Working With Database in NodeJS				✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	2	2	-	-	-	-	-	-	-
CO2	2	-	-	-	-	-	-	-	1
CO3	-	2	3	-	-	-	2	-	-
CO4	-	2	3	-	-	-	-	-	1
CO5	2	-	2	2	-	-	3	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA854 : Mobile Application Technology : Android Platform

(200 Marks)

Contact Hours: 06

Pre-requisite: Object oriented concepts and Java Programming.

Instructional Methods & Pedagogy: During theory lectures illustrations emphasizing the need for basic features of Mobile Computing and Android- the Mobile Application Development platform will be given. During Practical sessions, students will be required to develop Mobile Application using JAVA programming language in Android. Student shall also develop applications with elegant user interface that deal with data storage, documents sharing among applications and application based on Web Service and Google maps.

Outline of the Course:

<table>
<tr><th rowspan="2">Unit No.</th><th rowspan="2">Title of the Unit</th><th colspan="2">Minimum Numbers of Hours</th></tr>
<tr><th>Theory</th><th>Practical</th></tr>
<tr><td>1</td><td>Introduction Android and Android Development tools</td><td>7</td><td rowspan="6">36</td></tr>
<tr><td>2</td><td>User Interface in Android</td><td>8</td></tr>
<tr><td>3</td><td>Material Design</td><td>5</td></tr>
<tr><td>4</td><td>Data Handling in Android</td><td>7</td></tr>
<tr><td>5</td><td>Web Service and Android APIs</td><td>5</td></tr>
<tr><td>6</td><td>Cross Platform Mobile Application Development</td><td>4</td></tr>
</table>

Total Hours (Theory): 36

Total Hours (Lab): 36

Total: 72

Detailed Syllabus:

Unit - I: Introduction Android and Android Development Tool Hours: 07

- Introduction to Mobile Applications and Android
- Setting up development environment, Dalvik Virtual Machine, Android Application Structure, DDMS, Android Manifest File, Gradle, Android permissions, APK
- Basic Building blocks – Activities, Services, Broadcast Receivers & Content providers
- Basic Views: Toast, TextView, EditText, Button

- Components for communication -Intents & Intent Filters
  - Intents: Explicit Intents, Implicit Intents, Switching between activities and passing data between activities using Intents.
- Logging

Unit - II: User Interface in Android Hours: 08

- Working with Containers: LinerLayout, RelativeLayout, TableLayout, ScrollView, and FrameLayout, Fragment
- Views: ImageButton, CheckBox, ToggleButton, RadioButton, RadioGroup , ProgressBar, AutoComplete TextView
- Picker Views: TimePicker and DatePicker views
- Adapters: ArrayAdapter, BaseAdapter
- ListView and ListActivity, GridView, Gallery
- Menus: Option menu, Context menu, Sub menu
- Tabs and TabActivity
- Alert Dialogs
- Navigation Drawer
- Android Toolbar

Unit - III: Material Design Hour: 05

- Android Design Support Library
- Vector Drawables
- Basics of material design components and principles
- Android styles and themes
- Animations and Transitions

Unit - IV: Application Data Handling   Hour: 07

- Basics of Storage in Android
- Shared Preference : Save key-value data
- Store Data in Database
- Save files on Device
- Sharing of data
- Use of NFC for sending data

Unit - V: Web Services Hour: 05

- Working with Web Service in Android
- Deploying Web Service

Unit – VI: Cross Platform Mobile Application Development Hour: 04

Introduction to cross-platform development with PhoneGap. Native vs. Hybrid Development.
Introduction to PhoneGap: Understanding phonegap basics, understand architecture of phonegap application, learn how to configure your development environment, learn how to build your application without environment configuration, Build your application without mac.

HTML & CSS & JavaScript: Introduction to HTML5, CSS3, JavaScript, HTML5 tags overview, HTML5 for mobile development & it's advantages, JavaScript Overview, JQuery
PhoneGap Events: Basics of events, learn how to work with PhoneGap events

Core Books:

1. Wei-Meng Lee: Beginning Android 4 Application Development, Wiley India Pvt Ltd.
2. Reto Meier: Professional Android 2 Application Development, Wrox publication
3. Mark L. Murphy: The Busy Coder's Guide to Android Development

Reference Books:

1. Dawn Griffiths, David Griffiths: Head First Android Development, O'REILLY publication
2. Mark L. Murphy: Beginning Android 2, APRESS publication
3. Mark L. Murphy : The Busy Coder's Guide to Advanced Android Development, Commons Ware, LLC

Web References:

1. https://www.tutorialspoint.com/android/ [Android Tutorials]
2. https://developer.android.com/guide/ [Android Tutorials]
3. www.vogella.com/tutorials/android.html [Android Practical Tutorials]
4. https://www.androidhive.info/ [Android Practical Tutorials]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Be familiar with Android OS
C02 : Be familiar with Android Development Tools
C03 : Be able to create proper look and feel of mobile application
C04 : Be able to create interactive mobile application that handle data
C05 : Be able to create mobile application that works web service and Google Map.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction Android and Android Development tools		✓			
2	User Interface in Android		✓	✓		
3	Material Design		✓	✓		
4	Data Handling in Android		✓	✓		
5	Web Service and Android APIs		✓	✓	✓	✓
6	Cross Platform Mobile Application Development				✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	1	-	1	-	2	-	-	-	1
CO2	-	2	2	1	-	-	-	2	-
CO3	-	1	3	-	-	3	2	-	-
CO4	-	-	-	-	3	-	-	-	2
CO5	2	-	-	3	-	-	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA855 : Mobile Application Development : iOS Platform

(100 Marks)

Contact Hours: 06

Pre-requisite: Object oriented concepts and Basics of programming language, DBMS.

Methodology & Pedagogy: During theory lectures illustrations emphasizing the need for basic features of Mobile Computing and iOS - the Mobile Application Development platform will be given. During Practical sessions, students will be required to develop iOS based Mobile Application using SWIFT programming language. Student shall also develop applications dealing with data storage, XML parsing and JSON parsing.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	Practical
1	Introduction to MAC OS and Hybrid Mobile Application Platform	04	36
2	Introduction to Development Environment	07	
3	Introduction to Swift Programming Language	07	
4	Working with iOS App development paradigms	09	
5	Database Management in iOS	05	
6	Publishing App and Networking in iOS	04	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total: 72

Detailed Syllabus:

Unit – I: Introduction to MAC OS and Hybrid Mobile Application Platform Hours: 04

Mac OS X — An Overview, History of OS X, Main characteristics of mobile apps, Differences between mobile apps and desktop apps, How iOS is tailored to a mobile platform, iOS main components and services.

Introduction to Mobile Application Development Ecosystems, Introduction to Hybrid Platform Mobile Application Development.

Unit – II: Introduction to Development Environment Hours: 07

Introduction to XCODE, iOS App architecture, iOS developer center, App life cycle, interface builder, iOS simulator, creating the first project.

Unit – III: Introduction to Swift Programming Language Hours: 07

Overview of Swift, Swift environment, Basic syntax, Data types, Variables, Optionals, Tuples, Constants, Literals, Operators, Decision making, Loops, String, Array, Dictionaries, Functions, Structures, Class, Methods, Inheritance, Type casting, Protocols.
Apple Guidelines for releasing an ios app

Unit – IV: Working with iOS App development paradigms Hours: 09

Auto Layout, Views, Outlets and Actions, storyboard.

Different View Controller: single view Controller, Master-Detail View Controller, Navigation View Controller, Application delegate.
UI Controllers: Label, Button, Text Field, Slider, Switch, Progress View, Page Control, Table View, Collection View, Image View, Text View, Web View, Map View, Date Picker, Picker View, Search Bar, Gestures, push notification, Image Picker.
UserDefault class.

Unit – V: Database Management in iOS Hours: 05

Introduction to Core Data, Core Data stack, Defining Object Model, saving to core data, fetching from core data.
Introduction to SQLite, CRUD application using SQLite.

Unit – VI: Publishing App and Networking in iOS Hours: 04

Submit an iOS app to the App Store, Apple Developer Program, XML Parsing, JSON Parsing, Alamofire, Introduction to Cocoapods, installing pods in project.

Core Books:

1. Paris Buttfield-Addison, Tim Nugent : “Learning Swift: Building Apps for macOS, iOS, and Beyond, 3rd Edition”, Jonathon Manning, May 2018.
2. Abhishek Mishra : “Swift iOS Programming”, Wiley India Publications, 2016.
3. Matt Neuburg : ”iOS 11 Programming Fundamentals with Swift, 4th Edition”, O’Reilly, October, 2017.

Reference Books:

1. Boisy G. Pitre : “Swift for Beginners: Develop and Design”, Paperback, December, 2014.
2. Wei-Meng Lee : ”Beginning Swift Programming”, Paperback, February, 2015.

Web References:

1. https://www.macforbeginners.com/osx-guide/mac-os-x-introduction/ [Introduction to MAC OS].
2. https://codewithchris.com/xcode-tutorial/ [Introduction to Xcode].
3. https://www.tutorialspoint.com/swift/index.htm [Swift Programming Language] 4. https://www.raywenderlich.com/173972/getting-started-with-core-data-tutorial-2 [Example of Core Data].
5. https://www.raywenderlich.com/167743/sqlite-swift-tutorial-getting-started [Example of SQLite].

6. http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.453.6679&rep=rep1&type=pdf [Mobile Application Development Ecosystems].
7. https://developer.apple.com/app-store/review/guidelines/ [Apple Guidelines for App Publishing].

Course Outcomes: Upon successful completion of the course, the students will :

C01 : Be familiar with basics of an iPhone App and Hybrid Mobile App.

C02 : Be able to work with X-Code, and programming in Swift.

C03 : Be able to Design and Develop iphone applications.

C04 : Be familiar with Database Integration with App and Parsing of XML and JSON and extract data from it.

C05 : Be able to publish an iOS app to the AppStore.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	**C02**	**C03**	**C04**	**C05**
1	Introduction to MAC OS and Hybrid Mobile Application Platform	✓				
2	Introduction to Development Environment		✓	✓		
3	Introduction to Swift Programming Language	✓		✓		✓
4	Working with iOS App development paradigms			✓		✓
5	Database Management in iOS			✓	✓	
6	Publishing App and Networking in iOS				✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	1	-	1	-	2	-	-	-	1
CO2	-	2	2	1	-	-	-	2	-
CO3	-	1	3	-	-	3	2	-	-
CO4	-	-	-	-	3	-	-	-	2
CO5	2	-	-	3	-	-	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA856 : HTTP Web Service for Enterprise Application
(200 Marks)

Contact Hours: 06

Pre-requisite: Knowledge of C#,ASP.NET MVC, HTML, CSS, and have some understanding of JavaScript. Database connectivity knowledge.

Methodology & Pedagogy: During theory lectures illustrations emphasizing the need for advanced features of WEB API with ASP.NET will be given. During Practical sessions, students will be required to develop Web API using concepts discussed during class.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	Practical
1	Overview of ASP.NET Web API	07	36
2	Web API Controller and Model	06	
3	Web API Routing	05	
4	Web API Formatter and Exception Handling	05	
5	Hosting of Web API	05	
6	Managing Database and Consume Web API	08	

Total Hours (Theory): 36
Total Hours (Lab): 36
Total: 72

Detailed Syllabus:

Unit - I: Overview of ASP.NET WEB API Hours:07
Introduction to Service Oriented Architecture, Introduction to WEB API, Characteristics of ASP.NET WEB API, versions of Web API, Difference among web service, Window Communication Foundation and WEB API, Test Web API Fiddler and Postman

Unit - II: Web API Controller and Model Hours:06
Use of Web Api Controller in Controller class of Web API, Characteristics of Web API Controller, Difference between Web API and MVC controller, Action Method Naming Conventions, Action Result: void, HttpResponseMessage, IHttpActionResult and Other types, Use of Model in Web API, Validation in Model

Unit - III: Web API Routing Hours:05
Routing in Web API, Types of routing: Convention-based Routing and Attribute Routing, Routing and Action Selection and Attribute Routing

Unit - IV: Web API Formatter and Exception Handling   Hours:05

Data Formatter, Media Type Formatter, Web API Filter, Filter of Exception, HttpResponseException, Exception Filters, Registering Exception Filters, HttpError

Unit - V: Hosting of Web API and Securing Web API   Hours:05

IIS Hosting, Self-Hosting, hosting on the Open Web Interface for .NET, Hosting for Azure Mobile Services, Encrpty-Decrypt Text, Authentication: Filter, Basic, Forms and Integrate Window Authentication

Unit - VI: Managing Database and Consume Web API   Hours:08

CRUD operation with Entity Framework in Web API, Consume Web API through GET, POST, PUT and DELETE

Core Books:

1. Fillip Wojcieszyn: “ASP .NET WEB API 2 Recipes”, Apress, April 2011.
2. Uurlu, Ali, Zeitler, Alexander, Kheyrollahi, Ali: “Pro ASP.NET Web API HTTP Web Services in ASP.NET”, Apress, 2013.

Reference Books:

1. Akhil Mittal: “Diving into ASP.NET Web API”, 2016.
2. Ricardo Peres:” Entity Framework Core Cookbook”, 2nd Edition, Kindle Edition
3. Shankar Kambhampaty, “Service-Oriented Architecture: For Enterprise Applications”, WILEY India Edition

Web References:

1. http://www.tutorialsteacher.com/webapi/web-api-tutorials
2. https://www.udemy.com/learning-aspnet-web-api/

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Be familiar with SOA and technology to implement SOA in ASP.NET

C02 : Be aware about use of Model and Controller in Web API

C03 : Be familiar with Routing concept in Web API

C04 : Be able to use different formatter and exception handling in Web API

C05 : Be able to Host Web API using various ways

C06: Be able to manage database connectivity thorough Entity Framework

C07: Be able to consume Web API through different method of HTTP

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes						
		C01	C02	C03	C04	C05	C06	C07
1	Overview of ASP.NET WEB API	√						
2	Web API Controller and Model		√					
3	Web API Routing			√				
4	Web API Formatter and Exception Handling				√			
5	Hosting of Web API					√		
6	Managing Database and Consume Web API						√	√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	-	-	2	-	-	-	2	2
CO2	2	2	-	-	-	-	3	-	-
CO3	-	-	2	3	-	2	-	-	-
CO4	3	3	2	-	2	-	2	2	2
CO5	-	3	-	2	3	3	-	-	3
CO6	-	2	-	3	-	-	2	2	-
CO7	-	2	-	-	-	3	3	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA857 : Frameworks and Applications

(200 Marks)

Contact Hours: 06

Pre-requisite: Enterprise Computing.

Methodology & Pedagogy: During theory sessions the students shall be introduced to various frameworks. Details of Spring and Hibernate frameworks will be discussed and their integration to develop real world applications will be demonstrated. During practical sessions students will be trained to develop various standalone and web applications using the studied frameworks.

Learning Outcomes: Upon successful completion of the syllabus students shall be able to acquire in depth knowledge of frameworks and develop applications using the same. Students shall be having understanding of major concepts like DI, AOP, Web MVC, Spring - Hibernate Integration and HQL and will be able to identify its usage and apply them as per the need while developing applications.

Outline of the Course:

Unit No	Title of the Unit	Minimum Numbers of	
		Theory	Practical
1	Introduction to Spring	04	36
2	Beans and Containers	07	
3	The Application Context, Data Validation and Conversion	06	
4	Aspect-Oriented Programming	06	
5	Spring and Persistence	09	
6	Spring Web MVC and Spring Boot	04	

Total Hours (Theory): 36
Total Hours (Lab): 36
Total: 72

Detailed Syllabus:

Unit - I: Introduction to Spring Hours: 04

Introduction, Characteristics of framework, Types of framework(Existing frameworks),What is Spring?, The Spring Architecture, Overview of the Spring Modules, Spring Configurations, Wiring Bean, A Simple Example, Java Application Vs Spring Application

Unit - II: Beans and Containers Hours: 07

Spring Containers, Spring Configuration File, Spring Beans, Using the Container, The BeanFactory Interface, Singleton vs. Prototype, Bean Naming, Dependency Injection, Setter Injection, Constructor Injection

Unit- III: The Application Context Hours: 06

The ApplicationContext Interface, Accessing Application Components, Accessing Resources, Internationalization with MessageSource.

Unit - IV: Aspect-Oriented Programming Hours: 06

AOP Concepts, Join Points, Point Cuts, Advice, Configuration of Aspects - Types of Advice,AOPExample.

Unit - V: Spring and Persistence Hours: 09

Working with the HSQLDB Database, Integration with JDBC, Use of JdbcTemplate Class, Exception Translation, Updating with the JdbcTemplate Queries using the JdbcTemplate, Mapping Results to Java Objects.

Introduction to Object Relational Mapping, What is Hibernate?, The HibernateTemplate class, Hibernate Configuration Files, Mapping Classes and Fields for Hibernate, Creating and Saving a New Entity, Locating an Existing Entity, Updating an Existing Entity, Hibernate Sessions, Hibernate Query Language, Executing Queries, Hibernate annotation.

Unit - 6: Spring Web MVC and Spring Boot Hours: 04

What is Spring Web MVC?, Setting Dispatchers, Loading Configuration Files, Writing a Controller, Types of Controller, Configuring the Controller, Setting of Handler Mapping, Handler Mapping Options, Handling a Form

Integrating Hibernate with Spring MVC – Accessing Database, Storing Form Values and Retrieving Data from Database.

Introduction to Spring Boot - Spring Boot Features, Spring Boot Application.

Core Books:

1. Craig Walls, Ryan Breidnbach: Spring in Action, 3rd Edition.
2. Rod Johnson, Juergen Hoeller, Alef Arendsen, Thomas Risberg, Colin Sampaleanu: Professional Java Development with the Spring Framework.

Reference Books:

1. Rod Johnson: J2EE Applications Without EJB, Wiley Publication.
2. API Documentation (http://www.springsource.org/spring-framework#documentation).

Web References:

1. http://static.springsource.org/spring/docs/3.0.x/spring-framework-reference/html/overview.html[Spring framework docs]
2. http://www2.parc.com/csl/groups/sda/publications/papers/Kiczales-ECOOP97/for-web.pdf[Aspect Oriented Programming]
3. http://netbeans.org/kb/docs/web/quickstart-webapps-spring.html[Spring Web MVC]
4. https://www.javatpoint.com/spring-boot-tutorial [Introduction to Spring Boot].

Course Outcomes: Upon successful completion of the course, the students will :

C01 : Be familiar with basics of Frameworks and advantages of frameworks.

C02 : Be able to work with Dependency Injection and IOC .

C03 : Be able to use ApplicationContext to achieve DI.

C04 : Be familiar with AOP fundamental.

C05 : Be able to integrate Spring application with database using JdbcTemplate and Hibernate.

C06 : Be able to develop Spring Web MVC applications and Simple Spring Boot application.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes					
		C01	**C02**	**C03**	**C04**	**C05**	**C06**
1	Introduction to Spring	✓					
2	Beans and Containers		✓				
3	The Application Context, Data Validation and Conversion			✓			
4	Aspect-Oriented Programming				✓		
5	Spring and Persistence					✓	
6	Spring Web MVC and Spring Boot						✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	3	3	2	2	2	1	2	2
CO2	3	3	3	2	2	2	2	2	2
CO3	3	3	3	2	2	2	3	2	2
CO4	3	3	3	2	2	2	3	2	2
CO5	3	3	3	2	2	2	1	2	2
CO6	3	3	3	2	2	2	1	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA858: Multi Paradigm Scripting using Python

(Marks 200)

Contact Hours: 06

Pre-requisite: Programming principals and Logic Development, Fundamentals Concepts of Programming Language, Object Oriented Programming Using C++

Methodology & Pedagogy: Theory sessions are required to address computational power of python through its ability to deploy programs using functional, object oriented and web based aspect. Practical sessions demonstrate the implementation of the concepts which are taught during the theory sessions. Case study will help the students to come out with one working module in any of the paradigm through python.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1	Introduction to Python in context of Multi Paradigm Script	08	36
2	Getting Started with Python	08	
3	Python Objects, Mapping, Set & Iterative Programming	05	
4	Python functional Programming	05	
5	Object Oriented Programming using Python	05	
6	Advanced Programming using Python	05	

Total Hours (Theory): 36

Total Hours(Lab): 36

Total Hours: 72

Detailed Syllabus:

Unit – I: Introduction to Python in context of Multi Paradigm Script Hours: 08

Overview of programming Paradigms-Imperative, Functional, logic and object oriented Introduction to Python Programming Language, Downloading and Installing, Running Python, Python Documentation. Features of Python and its origin.

Unit – II: Getting Started with Python Hours: 08

Input through standard input device, Generating Output on console, Operators in python, how to use Numbers in python, String and its variations in python, Comment statement in python ,Various Statement & its Syntax, Python Identifiers, Basic Programming Style of Python, Memory Management scheme adopted in Python, Conditional & Branching Statement in python.

Unit – III: Python Objects, Mapping, Set & Iterative Programming: Hours: 05

Python objects and its variations: List, Tuple & Set
Different looping constructs available in Python: While, for, continue, break and pass statement Special Python object like Dictionary, nesting of one object into another/same object, and other built in functions for list, tuple, set and dictionary.

Unit – IV: Python functional Programming: Hours: 05

What are functions, calling functions, creating functions, passing functions, formal arguments, variable length arguments, default arguments, returning values from the functions, returning multiple values from the functions, functional programming, variable scope and recursion.

Unit – V: Object Oriented Programming using Python: Hours: 05

Basics of class, object and instance. Class level attribute and instance level attribute. Constructor and other magic methods. Bound and unbound methods. Built in functions for python class and objects.

Unit – VI: Advanced Programming using Python: Hours: 05

Class mappings, inheritance, Introduction to client server programming. Introduction to Python GUI programming using Tkinter

Core Books:

1. Wesley J. Chun : Core Python Programming, 2nd edition, Pentice Hall,2006.
2. Megnus Lie Hetland : Beginning Python from novice to professional, 2nd edition, Apress,2009.

Reference Books:

1. Mark Lutz : Programming Python, 4th Edition,O'reilly, 2011 .
2. Dusty Philips: Python 3 Object oriented Programming , PACKT publishing, 2010.
3. Steve Holden: Python Web Programming, 1st edition,2002.

Web References:

1. http://people.cs.aau.dk/~normark/prog3-03/html/notes/paradigms_themes-paradigmoverview-section.html [Overview of Programming Paradigms]
2. www.tutorialspoint.com/python [For tutorials of Python]
3. https://docs.python.org [Python Documentation]
4. https://www.python.org/about/gettingstarted/[For beginners of Python]
5. http://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-189-a-gentleintroduction-to-programming-using-python-january-iap-2008/ [Python Materials]

Course Outcomes: Upon successful completion of the course, the students will :

C01 : Install and run the Python interpreter, Create and execute Python Program

C02 : Adequately use standard programming constructs: repetition, selection, functions, composition, modules, aggregated data (arrays, lists, etc.)

C03 : Explore Python's object-oriented features

C04 : Able to use library software for building a graphical user interface or web application

C05 : Write Python functions to facilitate code reuse

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to Python in context of Multi Paradigm Script	✓				
2	Getting Started with Python	✓	✓			
3	Python Objects, Mapping, Set & Iterative Programming		✓			
4	Python functional Programming		✓			✓
5	Object Oriented Programming using Python			✓		
6	Advanced Programming using Python				✓	

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	1	-	2	-	-	-	-	-	2
CO2	-	1	3	2	-	-	-	-	-
CO3	-	3	-	2	-	1	1	-	-
CO4	-	-	-	3	-	-	2	1	-
CO5	3	1	2	-	1	-	1	-	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS705 (C): ACADEMIC WRITING

I. Credits and Schemes:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
II	HS705	Academic Writing	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title / Topic	Classroom Contact Hours
1	**Academic Writing and Research Process** • *Introduction to Academic Writing* • *Academic Writing as a Part of Research* • *Types of Academic Writing* • *Features of Academic Writing* • *Importance of Good Academic Writing in various Academic Works*	**05**
2	**Anatomy of Academic Writing** • *Academic Vocabulary* • *Simple and Complex Sentences* • *Organizing Paragraphs* • *The Writing Process* • *Adopting Academic Writing Style*	**05**
3	**Key Academic Skills** • Note – taking • Note – making • Paraphrasing • Summarizing	**05**
4	**Accuracy in Academic Writing** • *Lexical Range* • *Academic Language and Structures* • *Elements of Writing* • *Proof Reading, Editing, and Rewriting*	**05**
5	**Using and Citing Sources of Ideas**	**05**

	• *Academic Texts and their Types* • *Intellectual Honesty in Academic Writing* • *Avoiding Plagiarism – Idea Theft* • *Degrees of Plagiarism* • *Types of Borrowing* • *Anatomy of Citations* • *Common Citation Styles*	
6	**Contemporary Practices in Academic Writing** • Analytical Essays • Graph / Table / Process Interpretation and Description • Writing Reports • Writing Research / Concept Papers	**05**
	Total	**30**

III. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like writing, group work, independent and collaborative research, etc.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Paragraph Writing	1	3	03
2	Note-taking / Note-making	1	3	03
3	Paraphrasing / Summarizing	1	4	04
4	Essay Writing	1	5	05
5	Concept Paper Writing	1	10	10
5	Attendance and Class Participation			05
			Total	**30**

External Evaluation

The University Practical Examination will be for 70 marks and will test the professional communication skills and academic writing skills of the students.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical /Quiz/ Project / Academic Writing	-	70	70
			Total	**70**

Course Outcome (COs):
After completion of the course, the student would:

CO1 have sound understanding of the concept and applications of academic writing
CO2 have acquired enough knowledge of academic writing style, strategy and approach
CO3 be able to demonstrate error free and effective academic writing
CO4 be able to demonstrate ability to work on project/report/paper writing
CO5 understand the concept of plagiarism and learn to use different citation styles as a part of referencing

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	3	-	-	-	-
CO3	-	-	-	-	3	-	-	-	-
CO4	-	-	-	-	3	2	-	-	-
CO5	-	-	-	-	3	-	-	-	2

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

V. Reference Books / Reading

Essential Reading for Concepts
- Academic Writing for International Students, Routledge
- Academic Writing: A Guide for Management Students and Researchers. Monipally, M. M. & Pawar, B. S. Sage. 2010. New Delhi

Essential Reading for Activity and Teacher Resource
- ***Effective Academic Writing Level - 1,2,3,4 (Second Edition) By:*** Alice Savage, Patricia Mayer, Masoud Shafiei, Rhonda Liss, & Jason Davis; *Publisher: Oxford*

Additional Reading
- Writing Your Thesis (2^{nd} Edition) by Paul Oliver, Sage
- Development Communication In Practice by Vilanilam V J, Sage
- Intercultural Communication by Mingsheng Li, Patel Fay, Sage
- www.owl.perdue.edu

OBJECTIVES, TEACHING SCHEME & DETAILED SYLLABUS

FOR

MCA PROGRAMME
(3rd SEMESTER)

EFFECTIVE FROM
ACADEMIC YEAR 2019-20

Teaching and Examination Scheme (MCA Programme)

Effective from Year 2019-20

Semester-III

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA926	Open Source Frameworks	3	3	6	6	10	20	70	15	15	70	200
CA927	Data Analytics	3	3	6	6	10	20	70	15	15	70	200
CA928-CA931	Elective-IV	3	-	3	3	10	20	70	-	-	-	100
CA932	Mini Project	-	15	15	15	-	-	-	100		200	300
		9	21	30	30	300			500			800

Elective-IV

1. CA928 Artificial Intelligence
2. CA929 Digital Image Processing
3. CA930 Compiler Construction
4. CA931 Analysis and Design of Algorithms

CA926-Open Source Frameworks

(200 Marks)

Contact Hours: 06

Pre-requisite: CA850: Web Application Development using Open Source Technology

Methodology & Pedagogy: In theory sessions, the emphasis will be given on introduction, installation and configuration of laravel framework, routing and artisan in laravel, working with blade template and SQL interaction for database transaction, object relational mapper of eloquent ORM and form validations. During practical sessions, students shall implement the concepts taught in theory sessions.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1	Bootstrap Basics	05	36
2	Introduction to Laravel with Installation & Configuration	06	
3	Routing in Laravel & Introduction to Artisan	06	
4	Blade Template and SQL Interaction	08	
5	Eloquent ORM	06	
6	Form Validations	05	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total: 72

Detailed Syllabus:

Unit – I: Bootstrap Basics   Hours: 05

Introduction: File Structure, Basic HTML Template, Global Styles, Default Grid System, Basic Grid HTML, Offsetting Columns, Nesting Columns, Fluid Grid System, Container Layouts, Responsive Design, Introduction to Responsive Design.
Implementation: Typography, Code, Tables, Forms, Buttons, Images, Icons, Glyphicons, ropdown Menus, Button Groups, Button with Dropdowns, Navigations, Navbar, Breadcrumb, Pagination, label, badges, Typographic elements, thumbnails, alerts, progress bar, wells.

Unit – II: Introduction to Laravel with Installation & Configuration   Hours: 06

What is Laravel, features, MVC architecture, structure of laravel application (laravel directory structure), Basic requirements for Laravel, Using Laravel Installer, Using Composer, Linux & Windows, Finding and installing new packages

Introduction, Environment configuration, Protecting sensitive configuration, Maintenance mode, database configuration (setting database connection parameter for laravel and artisan)

Unit - III: Routing in Laravel & Introduction to Artisan   Hours: 06

Basic Routing, Route Parameters, Route Filters, Named Routes, Route Groups, Sub-Domain Routing, Route Prefixing, Route, Model Binding, Throwing 404 Errors, Routing to Controllers

Artisan Command Line Tool, database creation, artisan migration, migration structure, creation migration, Database seeding

Unit – IV: Blade Template and SQL Interaction   Hours: 08

Template inheritance, Master layout, Extending the master layout, display variables, Blade conditional statements, Blade Loops, Executing PHP functions in blade

Introduction, Running Raw SQL Queries, Database Transactions

Unit – V: Eloquent ORM   Hours: 06

Eloquent ORM Models: Naming conventions, table name & primary keys, timestamps

Basic Operations: Create, Retrieve, Update, Delete Using Models, displaying data from models in views.

Unit VI: Form Validations   Hours: 05

Defining The Routes, Creating The Controller, Writing The Validation Logic, Displaying The validation Errors, Array validations, creating new validators, Error messages & custom errors

Available Validators: Accepted, After (Date), Alpha, Alpha Dash, Alpha Numeric, Array, Before (Date), Between, Boolean, Date, Date Format, Different, Digits, Digits Between, E-Mail, Exists

(Database), Image (File), In, Integer, Max, Min, Not In, Numeric, Regular Expression, Required, String Custom validation rules.

Core Books:

1. Jake Spurlock : Bootstrap, O'reilly, May 2013.
2. Martin Bean: Laravel 5 Essentials, Packet Publishing, April 2015.

Reference Books:

1. Sheikh Heera: Laravel Reference Guide, Packet Publication, June 2016.

2. Matt Stauffer, Laravel: Up and Running: A Framework for Building Modern PHP Apps, paperback, 2016.

Web References:

1. https://laravel.com/docs/5.2 [Online Laravel 5.2 Documentation]
2. https://code.tutsplus.com/tutorials/build-web-apps-from-scratch-with-laravel-the-eloquent-orm--net-25631 [For Eloquent ORM]
3. https://laravel.com/docs/5.7 [For Blade Template, Routing, Database Connectivity]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Understand the basics of bootstrap in web application designing.
C02 : Be able to configure and install the Laravel framework.
C03 : Able to create MVC based application using the Laravel framework.
CO4 : Understand Template integration and database communication in the web application development.
CO5: Be able to develop the responsive web applications with validations.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Bootstrap Basics	✓		✓		✓
2	Introduction to Laravel with Installation & Configuration		✓	✓		
3	Routing in Laravel & Introduction to Artisan		✓	✓	✓	
4	Blade Template and SQL Interaction		✓	✓		
5	Eloquent ORM			✓	✓	
6	Form Validations			✓		✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	-	-	-	2	-	-	-	2	2
CO2	2	2	-	-	-	-	3	-	-
CO3	-	-	2	3	-	2	-	-	-
CO4	3	3	2	-	2	-	2	2	2
CO5	2	3	-	2	3	3	-	-	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA927 Data Analytics

(200 Marks)

Contact Hours: 06

Pre-requisite: CA845: Advanced Database Technologies

Methodology & Pedagogy: : During theory lectures the emphasis will be given on the basics of data analytics and related tools and techniques. Students will be introduced to the concepts of data science, exploratory data analysis, supervised and unsupervised learning methods. Applications as well as research trends and future direction of data analytics will be discussed with the length. During the practical sessions, students will be introduced to tools of data analytics such as R and Weka. Students will be given appropriate case studies of data analytics to get the real time exposure of data analytics.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	**Practical**
1	Introduction of Data Analytics	06	**36**
2	Exploratory Data Analysis and the Data Science Process	06	
3	Supervised Learning Methods	07	
4	Unsupervised Learning Methods	07	
5	Applied Data Science	05	
6	Recent trends and future directions	05	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total: 72

Detailed Syllabus:

Unit – I Introduction of Data Analytics Hours: 06

What is Data Science? ,Big Data and Data Science hype ,Why now? , Datafication , Current landscape of perspectives , Skill sets needed

Unit – II Exploratory Data Analysis and the Data Science Process Hours: 06

Philosophy of EDA - The Data Science Process ,Statistical Inference , Populations and samples , Statistical modeling, probability distributions, fitting a model

Introduction to R ,Basic tools (plots, graphs and summary statistics) of EDA

Unit –III: Supervised Learning Methods Hours: 07

Supervised methods: Linear Regression ,Classification Trees, Random Forest, Neural Networks: Different Models like single and multi-layer perceptron, back propagation, Application

Unit –IV: Unsupervised Learning Methods Hours: 07

Clustering, Association Rule Mining, Dimensionality Reduction - Singular Value Decomposition - Principal Component Analysis

Unit – V: Applied Data Science Hours: 05

Time Series Analysis, Recommendation Systems, Mining Social-Network Graphs

Unit – VI: Recent trends and future directions Hours: 05

Data Visualization - Basic principles, ideas and tools for data visualization,Data Science and Ethical Issues - Discussions on privacy, security, ethics , Next-generation data scientists, Basics of Big Data analytics

Core Books:

1. Cathy O'Neil and Rachel Schutt: Doing Data Science, Straight Talk From The Frontline, O'Reilly. 2014.
2. Jure Leskovec, Anand Rajaraman, and Jeffrey David Ullman,Mining of Massive Datasets Cambridge University Press,2nd Edition, New York, NY, USA,2014.
3. Howard B. Demuth, Mark H. Beale, Orlando De Jess, and Martin T. Hagan, *Neural Network Design* , paperback USA, 2nd Edition,2014.

Reference Books:

1. Walpole, R. E., Myers, R. H., Myers, S. L., & Ye, K. , Probability & statistics for engineers & scientists ,9th edition, Prentice Hall,2012.
2. Haykin, S. S., Haykin, S. S., Haykin, S. S., & Haykin, S. S., Neural networks and learning machines Pearson, Vlolume 3,2009.
3. Mohammed J. Zaki and Wagner Miera Jr, Data Mining and Analysis: Fundamental Concepts and Algorithms.,Cambridge University Press. 2014.

Web References:

1. https://onlinecourses.nptel.ac.in/noc17_mg24/preview [Online Data Analytics Course]
2. https://www.itl.nist.gov/div898/handbook/eda/section1/eda11.htm [Exploratory Data Analysis Material]
3. https://datahoarder.io/Humble%20Bundle%20Books/Humble%20Book%20Bundle_%20Data%20Science%20presented%20by%20O_Reilly/doingdatascience.pdf [Data Science E-Book]

4. http://www.astro.caltech.edu/~george/aybi199/Donalek_Classif.pdf [Supervised and unsupervised methods tutorials]
5. https://yourstory.com/2017/12/data-analytics-future-trends/ [Future Trends of Data Analytics]

Course Outcomes: Upon successful completion of the course, the students will :

C01 : Describe what Data Science and Data Analytics are and the skill sets needed to be a data scientist.

C02 : Understand significance of exploratory data analysis in statistical and visualization aspects.

C03 : Understand and apply data analytics techniques such as supervised,unsupervised and EDA.

C04 : Understand the importance of analytics in applications and able to implement them using data science techniques.

C05 : Able to understand the recent trends and future directions of data analytics.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction of Data Analytics	✓				
2	Exploratory Data Analysis and the Data Science Process		✓	✓		
3	Supervised Learning Methods			✓		
4	Unsupervised Learning Methods			✓		
5	Applied Data Science				✓	
6	Recent trends and future directions					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	2	2	1	-	-	-	1	-
CO2	2	2	1	1	-	1	-	-	-
CO3	1	3	3	2	-	1	3	-	-
CO4	2	1	2	2	2	1	2	-	-
CO5	1	1	2	3	1	-	2	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA928: Artificial Intelligence
(100 Marks)

Contact Hours: 03

Pre-requisite: Basic knowledge of Searching algorithms.

Methodology & Pedagogy: The theory sessions will be focused on basics of Artificial Intelligence, problem solving paradigms, and search strategies. Areas of application such as Expert Systems, Software Agents, knowledge representation, natural language processing, expert systems will be explored.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours
		Theory
1	Introduction to Artificial Intelligence	06
2	Knowledge Based Systems	05
3	Structured Knowledge Representations	06
4	Fuzzy Logic	07
5	Expert Systems	06
6	Emerging Trends in Artificial Intelligence	06

Total Hours (Theory): 36
Total: 36

Detailed Syllabus:

Unit – I Introduction to Artificial Intelligence: 06
Concepts and definitions of Artificial Intelligence (AI) − Brief history of AI − AI and related fields − The AI problems and underlying assumptions.

Unit – II Knowledge Based Systems: 05
Knowledge and its Components, Categories, Structure and Examples of KBS, Knowledge Acquisition and it's techniques.

Unit –III: Structured Knowledge Representations: 06
Review of propositional and predicate logic; resolution and theorem proving; non-monotonic inference; probabilistic reasoning; Bayes theorem.

Unit –IV: Fuzzy Logic: 07
Introduction- Fuzzyfication and Defuzzyfication, Fuzzy Sets, Fuzzy Operators and Arithmetic, Membership Functions, Fuzzy Relation, Fuzzy Rule Based System.

Unit – V: Expert Systems: 06
Introduction − Representing and using domain knowledge, Knowledge acquisition and representation , General structure of Expert Systems, Expert System Shell, Advantages and disadvantages of Expert Systems

Unit – VI: Emerging Trends in Artificial Intelligence: 06
Introduction - pattern recognition and classification process, learning Classification patterns, Visual image understanding, image transformation, Preliminary concepts of parallel and distributed Artificial Intelligence, Artificial Neural Networks, general Delta Rule, Back Propagation.

Core Books:

1. E. Rich and Knight, “Artificial Intelligence”, 2 nd Edition, TMH.
2. S. J. Russel and P. Norvig, “Artificial Intelligence: A Modern Approaches”, Prentice Hall, 2010.

Reference Books:

1. Timothy J. Ross, “Fuzzy Logic with Engineering Applications”, Willey Publlication.
2. D.W.Patterson, “Introduction to AI and Expert Systems”, PHI.
3. Elias M. Awad & Hassan Ghaziri, “Knowledge Management”, Pearson Publication.
4. D. W. Rolston, “Principles of AI and Expert Systems Development”, Mc Graw Hill.
5. P. H. Winston, “Artificial Intelligence”, Addison Wesley.

Web References:

1. http://www.tutorialspoint.com/artificial_intelligence/ [Basics of AI]
2. http://intelligence.worldofcomputing.net/ai-branches/expert-systems.html [Expert Systems]
3. http://pages.cs.wisc.edu/~bolo/shipyard/neural/local.html [Neural Network]

Course Outcomes: Upon successful completion of the course, the students will :

C01 : Able to understand the concept of Artificial Intelligence.
C02 : Able to understand Knowledge Based Systems.
C03 : Able to learn various Knowledge Representation techniques.
C04 : Able to understand fundamentals of Fuzzy Logic.
C05 : Get familiarize with the concept of Expert System as Application of AI.
C06 : Able to understand advanced concepts in AI.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes					
		CO1	CO2	CO3	CO4	CO5	CO6
1	Introduction to Artificial Intelligence	✓					
2	Knowledge Based Systems		✓				
3	Structured Knowledge Representations			✓			
4	Fuzzy Logic				✓		
5	Expert Systems					✓	
6	Emerging Trends in Artificial Intelligence						✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	2	2	1	-	-	-	1	-
CO2	2	2	1	1	-	1	-	-	-
CO3	1	3	3	2	-	1	3	-	-
CO4	2	1	2	2	2	1	2	-	-
CO5	1	1	2	3	1	-	2	-	-
CO6	3	2	2	1	-	3	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA929: Digital Image Processing
(100 Marks)

Contact Hours: 03

Pre-requisite: Knowledge of basic programming.

Methodology & Pedagogy: The fundamental concepts of digital image processing concepts should be covered in the beginning lectures. Various image enhancement techniques should be explained in details. The methods of image segmentation, representation and compressions should be taught in-depth.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours
		Theory
1	Digital Image Fundamentals	05
2	Image enhancement	07
3	Image Restoration and Reconstruction	05
4	Image Segmentation	07
5	Image Representations and Recognition	06
6	Image Compressions	06

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit – I: Digital Image Fundamentals Hours: 05

Introduction, Origin, Component of image processing system, Human visual system, Image as a 2D data, Image representation, Gray scale and Color images, image sampling and quantization, Basic Relationships between Pixels. Statistical parameters, Measures and their significance, Mean, standard deviation, variance, SNR, PSNR.

Unit – II: Image Enhancement Hours: 07

Basic gray level Transformations, Histogram Processing, Spatial Filtering, Preliminary Concepts, Extension to functions of two variables, Image Smoothing, Image Sharpening, Frequency domain filtering, Homomorphic filtering

Unit – III: Image Restoration and Reconstruction: 05

Noise Models, Noise Reduction, Inverse Filtering, MMSE (Wiener) Filtering.

Unit – IV: Image Segmentation: Hours: 07

Edge based segmentation, Region based segmentation, Region split and merge techniques, Region growing by pixel aggregation, optimal thresholding, Hough Transform, boundary detection.

Unit – V: Image representation and Recognition Hours: 06
Boundary representation, Chain code, Polygonal Approximation, Signature, Boundary segments, Boundary description, Shape Number, Fourier descriptor, Moments, Regional Descriptors, Topological Feature, Texture, Pattern and Patterns classes, Recognition Based on matching.

Unit – VI: Image compression Hours: 06
Fundamental, Image compression model, Error free compression, Variable length coding, Bit plan coding, Lossless predictive coding, Lossy compression, Compression standard.

Core Books:

1. Rafael C. Gonzalez, Richard E. Woods: Digital Image Processing, Fourth edition, Pearson education, 2017

2. Anil K. Jain: Fundamentals of Digital Image Processing, First Edition, Pearson Education,2015

Reference Books:

1. Burger, Wilhelm, Burge, Mark J: Principles of Digital Image Processing, Springer, 2009.

2. Milan Sonka, Vaclav Hlavac, Roger Boyle: Image Processing, Analysis, and Machine Vision, CL Engineering,2007

Web References:

1. https://www.tutorialspoint.com/dip/ [Basics of Image Processing]
2. https://sisu.ut.ee/imageprocessing/book/1 [All concepts of digital image processing]
3. https://www.cs.auckland.ac.nz/courses/compsci773s1c/lectures/ImageProcessing-html/topic3.htm [Image segmentation]
4. http://www.eletel.p.lodz.pl/mstrzel/imageproc/enhancement1.PDF [Image enhancement]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Understand the concept of Explain various types of images and analyze various techniques for intensity transformation and spatial filtering with applications.

C02 : Construct, Differentiate and analyze filtering in frequency domain and spatial domain.

C03 : Understand and able to Examine most frequently used compression techniques and exemplify different Morphological operations in bio-medical images with application.

C04 : Learn various image segmentation techniques.

C05 : Be able to represent and recognize images using various methods.

C06: Be able to gain understanding of basic image compression techniques.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes					
		C01	**C02**	**C03**	**C04**	**C05**	**C06**
1	Digital Image Fundamentals	✓					
2	Image enhancement	✓	✓				
3	Image Restoration and Reconstruction			✓			
4	Image Segmentation				✓		
5	Image Representations and Recognition					✓	
6	Image Compressions						✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	3	3	3	1	1	2	1	2
CO2	3	3	2	2	1	1	2	1	2
CO3	3	2	3	2	1	1	2	1	2
CO4	3	2	3	2	1	1	2	1	2
CO5	3	2	3	2	1	1	2	1	2
CO6	3	2	3	2	1	1	2	1	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA930 Compiler Construction
(100 Marks)

Contact Hours: 03

Pre-requisite: Basics of Programming.

Methodology & Pedagogy: The theory sessions will cover the concept and importance of compiler, language processors, lexical analyzer, translation, various code optimization techniques to understand the working of any compiler in detail.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours
		Theory
1	Introduction to Compiler	07
2	Lexical Analyzer	05
3	Syntax Directed Translation	05
4	Run-Time Environments	05
5	Code Optimization	07
6	Code Generation	07

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit – I: Introduction to Compiler Hours: 07

Introduction to Compilers: Introduction, Language Processors, The Structure of a Compiler- Analysis and Synthesis Phases, Lexical Analysis – The Role of Lexical Analyzer, Input Buffering, Specification of Tokens using Regular Expressions, Recognition of Tokens using Finite Automaton, The Lexical-Analyzer Generator-LEX

Unit – II: Lexical Analyzer Hours: 05

Introduction, Context-Free Grammars, Top-Down Parsing: Recursive Descent Parsing, Predictive Parsing, LL(1) Grammars, Bottom-Up Parsing: Shift-Reduce Parsing, Operator Precedence Parsing, Introduction to LR Parsing- SLR,More Powerful LR Parsers- CLR and LALR, Using Ambiguous Grammars, The Parser Generator- YACC.

Unit – III: Syntax Directed Translation Hours: 05

Introduction, Syntax Directed Definitions, Evaluation Orders for SDD's, Applications of Syntax Directed

Translation. Intermediate Code Generation: Variants of Syntax Trees, Three-Address Code, Types and Declarations, Translation of Expressions, Type Checking

Unit – IV: Run-Time Environments Hours: 05

Storage Organization, Stack Allocation of Space, Access to Non-local Data on the Stack, Heap Management, Introduction to Garbage Collection

Unit – V: Code Optimization Hours: 07

Introduction, Basic Blocks and Flow Graphs, Optimization of Basic Blocks, Machine Independent Optimizations – The Principal Sources of Optimizations, DAG Representation of Basic Blocks

Unit – IV: Code Generation Hours: 07

Introduction, Issues in the Design of a Code Generator, The Target Machine, A Simple Code Generator, Peephole Optimization, Register Allocation and Assignment, DAG for Register Allocation.

Core Books:

1. Alfred V Aho, Monica S Lam, Ravi Sethi, Jeffrey D Ullman – Compilers:Principles, Techniques &Tools – Pearson Education, Fourth Edition, 2014
2. Leland L Bech, System Software: An Introduction to Systems Programming, Pearson Education Asia, 1997.
3. Kenneth C. Louden, Compiler Construction: Principles and Practice,Thompson Learning, 2003.
4. J.P. Bennet, Introduction to Compiler Techniques, Second Edition, Tata McGraw-Hill, 2003.
5. Keith D Cooper and Linda Torczon, “Engineering a Compiler”, Morgan Kaufmann Publishers Elsevier Science, 2004.

Reference Books:

1. Mauro Pezze, Michael Young : Software testing and Analysis- Process, Principles and Techniques, Wiley India, 2012.
2. Dhamdhere D.M : Compiler Construction: Theory and Practice, McMillan India Ltd., 1983
3. Holub Allen : Compiler Design in C , Prentice Hall of India, 1990.

Web References:

1. http://www.sciencehq.com/computing-technology/compiler-construction.html [Introduction to Compiler Construction]
2. https://www.tutorialspoint.com/compiler_design/compiler_design_types_of_parsing.htm [Parsing]
3. https://www.geeksforgeeks.org/compiler-design-code-optimization/ [Code design and optimization]
4. https://cs.nyu.edu/courses/fall10/G22.2130-001/lecture12.pdf [Compiler Construction]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Understand basics of compiler construction.

C02 : Gain in-depth knowledge of lexical analyzer.

C03 : Understand basics of translation and runtime environments.

C04 : Learn code optimization.

C05 : Working knowledge of code generation.

Course Outcomes Mapping:

Unit No.	**Unit Name**	**Course Outcomes**				
		C01	**C02**	**C03**	**C04**	**C05**
1	Introduction to Compiler	✓				
2	Lexical Analyzer	✓	✓			
3	Syntax Directed Translation			✓		
4	Run-Time Environments			✓		
5	Code Optimization				✓	
6	Code Generation					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	2	2	1	1	-	-	1	-	2
CO2	2	2	2	1	-	-	1	-	2
CO3	2	2	3	1	2	-	1	-	2
CO4	2	2	1	1	-	-	1	-	2
CO5	2	2	3	2	-	-	1	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA931: Analysis and Design of Algorithms

(100 Marks)

Contact Hours: 03

Pre-requisite: Basic knowledge of Data Structures and Algorithms

Methodology & Pedagogy: During theory lectures the emphasis will be given on the basics of Data Structures and Algorithm Analysis. Students will be introduced with commonly used algorithms and methods like greedy method and dynamic programming. Backtracking and Branch and Bound methods are included.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours
		Theory
1	Basics of Algorithms and Data Structure	06
2	Divide and Conquer Algorithm	07
3	The Greedy Method	06
4	Dynamic Programming	06
5	Backtracking	06
6	Branch & Bound	05

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit – I Basics of Algorithms and Data Structure Hours: 06

What is an algorithm?, Algorithm Specification, Basics of Data Structure: Stacks, Queues, Linked Lists, Trees, Graphs, Heaps and Sets along with important operations on it, Concepts in Algorithm Analysis, Asymptotic Complexity

Unit – II Divide and Conquer Algorithm Hours: 07

The General Method, Binary Search, Finding Maximum And Minimum, Merge Sort, Quick Sort, Strassen's Matrix Multiplication

Unit –III: The Greedy Method Hours: 06

The General Method, The Knapsack Problem, Minimum-Cost Spanning Tree (Kruskal’s Algorithm, Prim’s Algorithm), Optimal Storage On Tape, Knap Sack, Job Sequencing With Deadlines Spanning Trees, Shortest Paths

Unit –IV: Dynamic Programming Hours: 06

The General Method, Multistage Graphs, Optimal Binary Search Trees, 0/1 Knapstack, Travelling Salesman Problem

Unit – V: Backtracking Hours: 06

Introduction, The Eight queens problem , Graph Coloring, Knapsack Problem, Hamiltonian Cycle

Unit – VI: Branch & Bound Hours: 05

The method (LC Search, 15 puzzle, FIFO Branch-and-Bound,LC Branch-and-Bound), Travelling Salesman Algorithm using branch & bound techniques

Core Books:

1. Horowitz Ellis & Sahni Sartaj : Fundamentals of Computer Algorithms, Galgotia Pub. Pvt. Ltd., New Delhi
2. Goodman, S. E. & Hedetnieni, : Introduction to the Design and Analysis of Algorithms, McGrawHill Book Comp.

Reference Books:

1. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein: Introduction to Algorithms, PHI.
2. Gills Brassard, Paul Bratley: Fundamental of Algorithms, PHI.
3. Anany Levitin: Introduction to Design and Analysis of Algorithms, Pearson
4. Dave and Dave: Design and Analysis of Algorithms, Pearson.

Web References:

1. http://www.digimat.in/nptel/courses/video/106106131/L01.html [Subject Overview]
2. https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-046j-design-and-analysis-of-algorithms-spring-2012/lecture-notes/[Detailing about all the algorithms]
3. https://www.tutorialspoint.com/design_and_analysis_of_algorithms/ [Detailing about all the algorithms]

Course Outcomes: Upon successful completion of the course, the students will :

C01 : Analyze the asymptotic performance of algorithms and data structure

C02 : Derive and solve recurrences describing the performance of divide-and-conquer algorithms

C03 : Understand and devise an algorithm Using the greedy method

C04 : Synthesize dynamic-programming algorithms, and analyze them and find out optimal solution by applying various methods

C05 : Understand the generic method for computing an optimal solution of a single-objective optimization problem

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	**C02**	**C03**	**C04**	**C05**
1	Basics of Algorithms and Data Structure	✓				
2	Divide and Conquer Algorithm	✓	✓			
3	The Greedy Method	✓		✓		
4	Dynamic Programming	✓			✓	
5	Backtracking	✓			✓	
6	Branch & Bound	✓				✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	2	2	1	1	-	-	1	-	2
CO2	2	2	2	1	-	-	1	-	2
CO3	2	2	3	1	2	-	1	-	2
CO4	2	2	1	1	-	-	1	-	2
CO5	2	2	3	2	-	-	1	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA932: Mini Project
(300 Marks)

Contact Hours: 15

Guidelines: Minor Project is in house project development. Every student is required to carry out Mini Project work under the supervision of a guide provided by the MCA Programme placement Coordinator. The guide shall monitor progress of the student continuously. A candidate is required to present the progress of the Mini Project work during the semester as per the schedule provided by the MCA Programme placement Coordinator.

Mini Project proposal should be prepared in consultation with project guide. It should clearly state the objectives and environment of proposed Mini Project to be undertaken. Project documentation must be with the respect to the project only. Project report should strictly follow the points suggested in format of project report. MCA programme placement coordinator will provide the format of project report. Student has to submit one copy of Mini Project to the institute. Each Student is required to make a copy of Mini Project in CD and submit along with Project report.

Course Outcomes :

CO1 : Student will understand the implementation of concepts of SDLC and Software Engineering.

CO2 :The programming concepts they learn during their academics, it will be converted in to the actual implementations.

CO3: Students will be exposed to understand the requirement of proposed software and implement these requirements in terms of programming logic and methods.

CO4: Students must understand the difference between a program and professional application/product/software.

CO5: Students will learn different categories of applications like Desktop application, Web applications, etc.

Evaluation: The project report shall normally be written in English in the specified format and shall be characterized by significant contribution to knowledge in the field. The Project report prepared according to approved guidelines and duly signed by the guide and the Head of the Department shall be submitted to the Head of the Institution. The evaluation scheme of Project is as under:

Course	Course Title	Teaching Scheme		Internal	End Semester Examination		Total
		Contact Hours	Credit	Continuous Evaluation	Report	Presentation & Viva	
CA932	Mini Project	15	15	100	100	100	300

The internal evaluation of project is done based on progress reports and internal presentations. The final evaluation of the project will be based on the project report submitted and a Viva Voce Examination by a Board of Examiners.
If a candidate fails to submit the project report on or before the specified deadline, he/she is deemed to have failed in the Project Work and shall re-enroll for the same in a subsequent semester. If a candidate fails in the viva-voce examinations of Project work he/she shall resubmit the project report within specified duration decided by university. The resubmitted project will be evaluated during the subsequent academic session. A copy of the approved project report after the successful completion of viva examinations shall be kept in the library of the college / institution.

Web References:

1. http://techwhirl.com/writing-software-requirements-specifications/[For effective SRS]
2. http://www.ibm.com/developerworks/websphere/library/techarticles/0306_perks/perks2.html [For best practices of Software Project Development]
3. http://www.uacg.bg/filebank/acadstaff/userfiles/publ_bg_397_SDP_activities_and_steps.pdf[Requirement analysis guidelines]
4. http://www.cs.wustl.edu/~schmidt/PDF/design-principles4.pdf [Software Design Principles and Guidelines]
5. http://www.cse.hcmut.edu.vn/~hiep/KiemthuPhanmem/Tailieuthamkhao/Effective%20Software%20Testing%20-2050%20specific%20ways%20to%20improve%20your%20testing.pdf[ForEffective Software Testing]
6. http://www.cs.uics.edu/~jbell/CourseNotes/OO_SoftwareEngineering/SE_Project_Report_Template.pdf [For guidelines to prepare software project report]

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	3	3	3	3	3	3	2	3
CO2	3	3	3	3	3	3	3	2	3
CO3	3	3	3	3	3	3	3	2	3
CO4	3	3	3	3	3	3	3	2	3
CO5	3	3	3	3	3	3	3	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

OBJECTIVES, TEACHING SCHEME & DETAILED SYLLABUS

FOR

MCA PROGRAMME
(4th SEMESTER)

EFFECTIVE FROM
ACADEMIC YEAR 2019-20

Semester-IV

Effective from Year 2019-20

Course Code	Course Title	Teaching Scheme				Internal	End Semester Examination		Total
		Contact Hours			Credit	Continuous Evaluation	Report	Presentation & Viva	
		Inst.	Industry	Total					
CA933	Project Work	2	28	30	30	200	200	400	800

CA933: Project Work

(800 Marks)

Contact Hours: 30

1. Project/dissertation work:

Students of MCA have to do the Project work in an Industrial/ Research Organization of computer field. Project work shall be carried out under the supervision of a qualified teacher in the Department as well as an expert from organization. Students have to meet the institute supervisor periodically and to attend the project/dissertation review meetings for evaluating the progress. The Project work shall be pursued for a minimum of 16 weeks during the semester.

1.1 MORI Principle to choose the topic:

Final semester project work is important and challenging element of MCA study. Selection of topic of project work is very crucial and important aspect of that. Following are important principles that will help students to select the appropriate topic of project work.

- **Manageable:** your project topic must be sufficiently focused so that it is possible for you to do the topic justice within the available time (one semester). You may have a real interest in, say, *'the impact of technology computers on Indian Economic Growth since Independence*, but you certainly won't be able to cover this topic in any detail in the space of one semester
- **Original**: this relates to the above point, since a topic that is focused and manageable is more likely to be one that has not been written about too extensively, thus leaving room for your original contribution. Ideally you will find an interesting and well-chosen topic which will impress those marking your work.
- **Relevant:** your project should clearly be relevant to some aspect of your studies, but it might also be relevant to your plans for, say, postgraduate study or a career. The dissertation may also be relevant in the sense that it plays to some of your established strengths, such as a particular course module or topic that you have enjoyed studying and in which you have previously done well.
- **Interesting:** you are obviously more likely to enjoy and be successful in your dissertation if it is of real interest to you and to those marking your work. Ask yourself if you are sufficiently committed to your idea to be able to give it your best throughout the duration of your project. You should also ascertain whether your supervisor finds the idea interesting during your initial discussions with her or him.

1.2 Evaluation of project/dissertation work:

The project report shall normally be written in English in the specified format and shall be characterized by significant contribution to knowledge in the field. Normally two copies of the report are to be submitted for evaluation. The Project report prepared according to approved guidelines and duly signed by the supervisor(s) and the Head of the Department shall be submitted to the Head of the Institution. The evaluation scheme of Project/Dissertation is as under:

Course	Course Title	Teaching Scheme				Internal	End Semester Examination		Total
		Contact Hours			Credit	Continuous Evaluation	Report	Presentation & Viva	
		Inst.	Industry	Total					
CA933	Project Work	2	28	30	30	200	200	400	800

The internal evaluation of project is done based on progress reports and internal presentations.

The final evaluation of the project will be based on the project report submitted and a Viva-Voce Examination by a Board of Examiners.

If a candidate fails to submit the project report on or before the specified deadline, he/she is deemed to have failed in the Project Work and shall re-enroll for the same in a subsequent semester. If a candidate fails in the viva-voce examinations of Project he/she shall resubmit the project report within specified duration decided by university. The resubmitted project will be evaluated during the subsequent academic session. A copy of the approved project report after the successful completion of viva examinations shall be kept in the library of the college / institution.

Course Outcomes :

CO1 : Student will understand the implementation of concepts of SDLC and Software Engineering.

CO2 :The programming concepts they learn during their academics, it will be converted in to the actual implementations.

CO3: Students will be exposed to understand the requirement of proposed software and implement these requirements in terms of programming logic and methods.

CO4: Students must understand the difference between a program and professional application/product/software.

CO5: Students will learn different categories of applications like Desktop application, Web applications, etc.

Web References:

1. http://www.microtoolsinc.com/Howsrs.php[For Software Requirement Specification Guidelines]
2. http://techwhirl.com/writing-software-requirements-specifications/ [For goals of SRS]
3. https://kepler-project.org/developers/reference/software-development-guidelines[For Software Development Guidelines]

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	3	3	3	3	3	3	2	3
CO2	3	3	3	3	3	3	3	2	3
CO3	3	3	3	3	3	3	3	2	3
CO4	3	3	3	3	3	3	3	2	3
CO5	3	3	3	3	3	3	3	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”